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0. Overview

The workflow on one page. Each panel is a stage of the pipeline and each is developed in the section named beside it. Lane A sets the acceptance criteria, B the data and the point-in-time screen, C the feature set, D the split, selection and tuning, E the assessment and the live book.

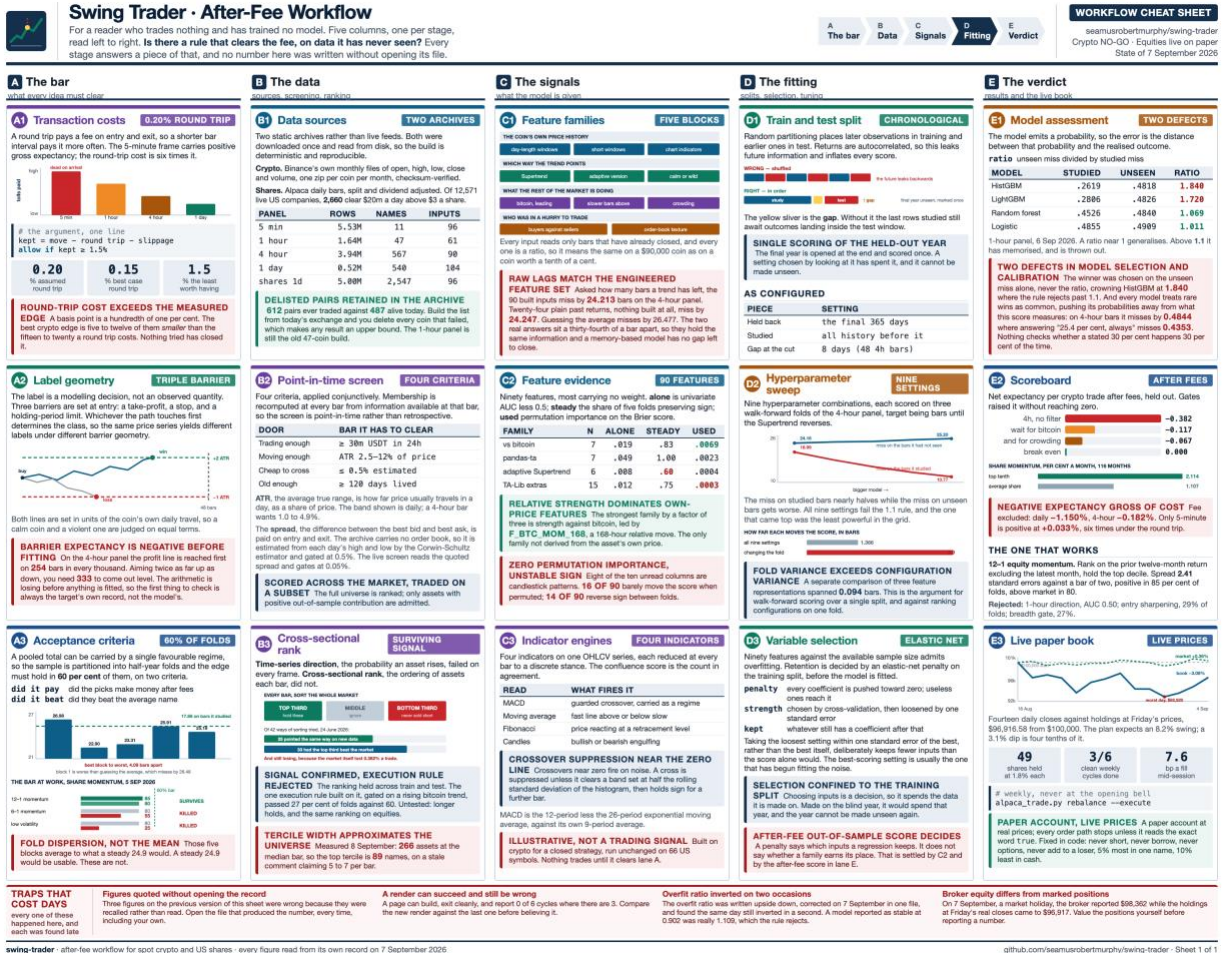


Fig X: Workflow cheatsheet showing key components and configuration steps, with panels A1 to E3 discussed in the following sections below. (Link to high-resolution cheatsheet).

0.1 After-Fee Evaluation

The goal is one trading strategy that makes money after fees on data it has never seen. Every stage below serves that test, and nothing is accepted without passing it.

A candidate is scored on a blind final year that is held out from all training and tuning. It must clear three bars at once: beat a coin flip at the real base rate, beat buy and hold, and return more than zero per trade after the round-trip fee of about 0.20 percent. A rule that ranks trades well but still loses to the fee is a NO-GO. The fee is the adversary, not the market.

The search runs across three decision frames so intraday, interday, and longer holding each get a fair test. A frame is a whole system: its own bars, its own label, its own features, scored on its own

out-of-sample. The frames are compared, never pooled, because a five-minute label and a four-hour label do not mean the same thing.

0.2 Pipeline Overview

The work splits into three parts that run in order. Each hands a clean input to the next.

Notebook	Role	The question it answers	Key outputs
Part 1, Metrics	Signal layer	What is worth measuring on a coin	indicators, entry and exit signals, a ranked shortlist, the decision checklist
Part 2, Controls	Risk and rules layer	What the account is allowed to do	position caps, stops, the daily loss circuit, the fee model, the four-gate screen, position sizing
Part 3, Execution	Modeling and search layer	Whether a tradeable edge survives fees	the datasets, the features, the model, and the after-fee scoreboard that returns GO or NO-GO

Metrics defines the signal vocabulary. Controls sets the constraints any strategy operates under, principally the transaction cost and the loss limits. Execution builds the datasets and the model and runs the out-of-sample test. The controls are not applied at the end; they are the constraint set the search is evaluated against.

0.3 Data Frames

Each frame is built from the same Binance Vision archives into its own dataset, `dataset_5m_allmarket`, `dataset_1h_allmarket`, `dataset_4h_allmarket`, and `dataset_1d_allmarket`, all stored as Parquet. One builder serves all three and retunes its feature windows, its label horizon, and its liquidity and volatility screens to the frame.

Frame	Bar	Trade style it targets	Default label
5m	5 minutes	intraday scalps	a shorter ATR barrier, roughly two hours
1h	1 hour	day-to-day swings	+2 / -1 ATR triple barrier, two wall-clock days
4h	4 hours	multi-day holds	+2 / -1 ATR triple barrier, two wall-clock days
1d	1 day	position swings, the fee-count lever	+2 / -1 ATR triple barrier, two daily bars

The higher-timeframe context shifts with the frame, so a five-minute model still sees the one-hour and four-hour trend, a one-hour model sees the four-hour and daily trend, and a four-hour model sees the daily and weekly trend. That is how each frame gets multi-resolution information without the datasets ever being merged. Every modelling section in Part 3 runs this same 1h, 4h, 5m loop

and reports one block per frame. The 1d frame was added 2026-08-17 as the fee-count lever: the survivorship-complete daily archive (669 pairs) is on disk and its baseline dataset and edge matrix are section 3.14's first read.

0.4 Equity Data

The second venue is Alpaca, a United States equity broker, and it carries the only strategy that has cleared the after-fee out-of-sample test. It is introduced alongside the Binance archives because the crypto search is not the whole project. The method was developed on crypto and first produced a positive result on equities.

The two data layers serve the same pipeline and differ in the respects below.

	Binance, crypto	Alpaca, US equities
Source	<code>data.binance.vision</code> monthly archives, checksummed, downloaded once	Alpaca's historical bars API on the SIP consolidated feed, split and dividend adjusted
Bars	5 minutes, 1 hour, 4 hours, 1 day	daily only
Span	longest available per pair, back to 2017 for BTC	2016 onward, the coverage floor of the feed
Universe	669 USDT pairs, including every pair ever delisted	2,660 names that pass a screen, all of them alive today
Delisted names	recovered, so estimates are unbiased	absent, so estimates are an upper bound
Trading hours	continuous	weekdays, 09:30 to 16:00 New York
Cost per round trip	0.15 percent engineered, 0.20 percent modelled	roughly 4 basis points measured, so about one twenty-fifth of the crypto cost

The cost row is the reason the equity track exists. Every crypto candidate was rejected on net expectancy rather than on discrimination. A venue with a round-trip cost roughly one twenty-fifth as large changes which effect sizes are economically actionable: an edge below a 0.20 per cent round trip may clear a 0.04 per cent one.

Survivorship bias and its direction. The Alpaca assets endpoint enumerates currently listed securities, so a firm delisted in 2019 is absent from the panel. Excluding failures biases realised performance upward, so every equity figure is an upper bound. This is the inverse of the Binance procedure, where the archive was crawled to recover delisted pairs. The bias is accepted for the first read and stated wherever a figure derived from the panel appears. `03-inputs/alpaca_data.py` writes the caveat into the universe snapshot itself, and every record built on the panel repeats it. A separate stress test, `03-inputs/equity_survivorship_stress.py`, injects delistings and re-scores, and the edge held.

How the data is built. Three commands, mirroring the crypto stages. `alpaca_data.py universe` lists every active, tradable name on NYSE, NASDAQ, ARCA, AMEX and BATS, then keeps those above a 20 million dollar median daily turnover and a 3 dollar price floor. On 26 August 2026 that took 12,571 names down to 2,660. `alpaca_data.py download` pulls maximum-history daily bars for those names into

one Parquet file per symbol, 4.1 gigabytes on disk, resumable, skipping anything already current. `build_dataset_equity.py` then runs the panel through the same builder the crypto frames use, with SPY standing in the market-factor seat that BTC occupies on the crypto side, and with the volatility screen widened to the 1 to 8 percent daily range equities live in. The result is `dataset_eq1d_allmarket.parquet`, 4,996,861 rows across 2,547 names, 96 features, built 18 August 2026.

Column naming under the shared builder. The equity panel retains the crypto column names, so `f_btc_rs` holds relative strength against SPY rather than against bitcoin. The names were preserved so that downstream code reads both panels without a mapping layer.

Result on the equity panel. The pre-registered monthly factor test, `03-inputs/equity_momentum_monthly.py`, ranks securities on the prior twelve-month return excluding the most recent month, holds the top decile for one month, and repeats on non-overlapping periods. No component is selected using this project's own model scores, so the test is out-of-sample by construction. Over 116 months to 5 September 2026, after a 5 basis point monthly cost, the top tenth beat the market average by 1.006 percentage points per month, with a t-statistic of 2.41, an absolute return above zero in 85 percent of half-year folds and a spread above zero in 80 percent of them. This is the only candidate in the project to clear the acceptance criteria. The same harness rejected the six-month formation and the low-volatility factor, consistent with the published literature, which establishes that the harness can return a negative result.

The most recent fold is the counterweight. In the first half of 2026 the top decile exceeded the market by 5.319 percentage points a month, the maximum in the series; over the three months since it has trailed by 5.646 points a month, the minimum. The paper book was funded during the second of these. Section 5 reports the realised effect.

Execution. `03-inputs/alpaca_trade.py` holds the fifty highest-ranked securities at 1.8 per cent of equity each and rebalances weekly. The long-only and no-margin constraints are enforced in code rather than by operator discipline. The account is a paper account and `LIVE_TRADING` remains false. Appendix D and section 5 report the running book.

0.5 Hard Rules

Every candidate strategy lives inside a fixed set of controls, summarised here and detailed in Part 2. None is negotiable, and the after-fee test in the last row overrides any in-sample result. The rows marked *revised* carry the adjustments logged below; the rest are unchanged from the charter.

Control	Setting	Why
Position cap	5 percent of equity at entry	one bad name cannot sink the book
Default size	half Kelly, a quarter on minimum signals	size to the edge, not the conviction
Fat-pitch exception	one position to 10 percent, with reward-to-risk at least 3:1, a named cause, and a written exit	rare, documented, reversible
Label geometry	longer-horizon, less fee-punishing triple barrier, replacing the +2 / -1 ATR default	the old label lost before any prediction; fewer round trips cut fee drag

Control	Setting	Why
Hard stop	ATR-scaled per frame, provisional, replacing the fixed 7 percent	trend protection sized to each coin's volatility, not a flat percent
Trailing stop	ATR-scaled per frame, provisional, replacing the fixed 10 percent	lets winners run scaled to volatility; settled by the exit-geometry sweep
Regime gate	act only when BTC is trending up	deploys the real but drowned cross-sectional signal in the regime where it works
Narrow book	entries only in coins that carried the out-of-sample gated edge and pass the live liquidity screen	a minority of coins carry all the profit; the average hides it
Fee tier	maker entries, BNB-discounted fees, measured from the account each run	the achievable 0.15 percent round trip closes half the gap to zero
Daily circuit	halt new orders if rolling 24-hour drawdown passes 3 percent	stop the bleeding, existing stops stay live
Drawdown ramp	below a 5 percent rolling-week loss, cut new size with each further 1 percent	shrink the book, do not just block it
Cash floor	keep at least 10 percent in cash	always able to act
Position limit	at most 3 new positions per week	forces selectivity
Direction	spot only, never short, never margin, never leverage	the mandate; shorting is logged below as research, not policy
Averaging down	never	a loser is exited or held, not fed
Anchoring	cost basis never enters hold or sell logic	decide on forward value only
The bar	nothing ships unless it beats a coin flip, buy-and-hold, and the fee out of sample	the fee is the adversary

Adjustments logged. Six changes were made to release model potential without adding ruin risk. All are bounded and stay inside the no-short mandate. They are documented here as the intended design; the code-level work lands in Part 3 (the label, exit, and regime sections) and has not yet been rebuilt into the on-disk datasets.

1. Exit geometry. The fixed 7 percent hard stop and 10 percent trailing stop become ATR-scaled per frame and stay provisional until the exit-geometry sweep, which fits per-coin trailing stops and a time-decaying take-profit and scores them after fees. The flat percent ignored each coin's volatility and conflicted with the code's existing 5 percent and roughly 8.5 percent ATR stops.
2. Label. The default +2 / -1 ATR triple barrier is replaced by a longer-horizon, less fee-punishing geometry with fewer round trips. The old label had negative unconditional expectancy, a base rate near 0.313 against a breakeven near 0.333, so it lost before the model predicted anything.
3. Regime gate Deploy the cross-sectional signal only when BTC is trending up. Relative strength is real and broad but drowned by a negative universe baseline, and a regime filter is the cheapest in-mandate way to test whether it turns positive.

4. Longer-horizon frame. Extend the search toward the daily frame alongside 4h, because coarser frames cut the fee count per unit of return, which is the central obstacle. As of 2026-08-17 the daily archive is complete and the baseline build is in flight (3.14).
5. Fee engineering (2026-08-17). The modelled cost gains a second, achievable scenario: post-only maker entries (no spread paid), BNB-discounted 0.075 percent per side, a 0.15 percent round trip (ACHIEVABLE_COST_PCT in 03-inputs/train_model.py). The live wrapper reads the account's actual maker and taker rates and BNB-burn setting at the start of every run (fee_status) and warns when the measured round trip exceeds the scenario; the paper book charges the same engineered rate.
6. Narrow book (2026-08-17). 03-inputs/narrow_book.py intersects the coins that carried the out-of-sample gated edge (positive contribution, at least three cohort appearances) with the live liquidity screen and writes 05-research/memory/narrow-book.json; 03-inputs/trade_binance.py refuses entries outside it. Rank broadly, trade narrowly.

Unchanged and kept tight. The 5 percent position cap, half-Kelly sizing, the 3 percent daily circuit, the drawdown ramp, the 10 percent cash floor, the three-new-per-week cap, no averaging down, no anchoring, and the after-fee bar. These bound survival, not the search. Loosening them would amplify a model that is still at the coin-flip floor, which is how no edge becomes ruin.

Open question on shorting. Going long the strong third and short the weak third would monetize the cross-sectional spread directly, even against a negative baseline. It is left out because it breaks the spot-only mandate and imports unbounded loss and liquidation risk. If ever tested, it should be small, market-neutral, unleveraged, and paper-first, still behind the after-fee bar.

0.6 Running It

The notebooks run on a MacPorts Python through a project .venv kernel. Secrets live in the run environment, never in the repository: the Binance and Alpaca keys, BINANCE_TESTNET, and the single money switch LIVE_TRADING, which stays false until a strategy has earned the right to trade. No real order is placed while it is false; the agent reads the market or runs against the Binance testnet.

One knob drives the modelling cells. LOAD_FRAME selects the active frame, 1h by default. Import Data loads that frame's Parquet into df and feat, and every modelling cell consumes those two. The multi-frame sections in Part 3, model assessment, stability, edge diagnostics, selectivity, and regime conditioning, loop over 1h, 4h, and 5m on their own and restore the active frame when they finish, so the rest of the notebook is unaffected.

Run order is top to bottom. The code cells carried into this document have their outputs cleared, so run them on the Mac with the .venv kernel to regenerate every figure and table. One environment caveat: the repository sits on an exFAT volume that scatters _ AppleDouble files, which can crash matplotlib; clear them if an import throws a 0xb0 decode error.

1. Trader Metrics

The signal layer, from Part 1 of this document. A read-only daily scan over public market data, no keys and no orders, sharing Part 3's data spine: price from the survivorship-complete data.binance.vision 1h archives resampled to daily, and the universe from Part 3's point-in-time Stage B screen.

The sections below present the signal families the model in Part 3 actually ranks and trades on, cross-sectional relative strength, the momentum family including MACD, the Supertrend and BTC-relative trend signals, the ATR-scaled exits, and the regime state, each tied to the feature it

becomes and the evidence for it. The runnable daily indicator board is kept throughout as an illustrative within-coin read, a place to eyeball candidates, not a trade trigger. Any signal must clear the after-fee, out-of-sample bar in Part 3 before it means anything, and on the 1h frame the verdict so far is NO-GO.

1.1 Environment Setup

The scan runs on Python 3.11 with dependencies pinned in 03-inputs/requirements.txt and reinstalled on each fresh kernel: pandas-ta-classic for indicators, pykalman for smoothing, plotly and matplotlib for charts. No exchange login or API key is needed, because both the price history and the universe come from Part 3's offline archives through build_dataset_1h. ccxt is kept only for an optional live top-up of the still-forming day.

Provenance is proven, not asserted. Price comes from the official data.binance.vision monthly per-symbol OHLCV zips, exchange-direct and checksummed, covering every USDT pair ever listed including delisted ones. That survivorship-complete source is what removes the universe-level bias the old live top-N-by-volume scan carried. The second cell pulls one month of BTCUSDT bars straight from the archive at 1h, 4h, and 5m to show the source and format.

```
Environment ready · python 3.12.13 (/Volumes/PortableSSD/Github/day-trader/.venv/bin/python)
ccxt 4.5.59 · pandas 2.3.3 · numpy 2.2.6 · plotly 6.8.0
```

frame	download command \
0 1h (old)	acquire_vision.py download --interval 1h
1 1d (old)	flow_data.py --interval 1d --all-market
2 4h (new)	acquire_vision.py download --interval 4h
3 5m (new)	acquire_vision.py download --interval 5m --sym...

raw klines on disk	built dataset
0 klines_1h/<SYM>/	dataset_1h_allmarket.parquet
1 klines/<SYM>/	daily_flow.csv
2 klines_4h/<SYM>/	dataset_4h_allmarket.parquet
3 klines_5m/<SYM>/	dataset_5m_allmarket.parquet

LIVE proof of source + format (one month of BTCUSDT, straight from the archive):

```
1h : 744 bars, 42 KB <- https://data.binance.vision/data/spot/monthly/klines/BTCUSDT/1h/BTCUSDT-1h-2024-01.zip
4h : 186 bars, 11 KB <- https://data.binance.vision/data/spot/monthly/klines/BTCUSDT/4h/BTCUSDT-4h-2024-01.zip
5m : 8928 bars, 455 KB <- https://data.binance.vision/data/spot/monthly/klines/BTCUSDT/5m/BTCUSDT-5m-2024-01.zip
```

1.2 Rank Universe

The decision-grade ranking in Part 3 is cross-sectional, not within-coin. At each bar it ranks the screened universe by a strength signal and holds only the top third, reading that cohort's after-fee return against the market and the bottom third (03-inputs/cross_sectional_4h.py). This is the classic momentum result: relative strength across coins often carries an edge where predicting each coin in isolation does not. The universe is thin, a median of five to seven coins per 4h bar, so it ranks into terciles at a five-coin floor rather than deciles.

The finding is real but drowned. Across the candidate signals most show the top third beating the market and a positive long-short spread that holds across train and test, the most robust being the adaptive-Supertrend direction and the daily Supertrend state. But the universe's own after-fee baseline is about minus 0.38 percent per trade, so even the best top third beats the market while staying below zero. The edge exists; the negative baseline sinks it. The levers to lift it are the regime gate, a less fee-punishing label, and a coarser frame, all logged in 0.5.

The runnable board below is the illustrative within-coin read. It loads 400 daily bars per coin from the 1h archives, computes a Kalman smoother, Bollinger Bands, Ichimoku, the Archer trend flag, RSI, Choppiness, and the triple-Supertrend, and ranks Supertrend-first. Stablecoins are dropped as flat by construction. A flagged cell is a candidate for review against the Part 3 evaluation, not a signal in itself.

```
Data spine: survivorship-complete 1h archives -> 1d bars | 400 bars/coin | board top 20 | live top-up False

Engine ready.

2026-09-08: survivorship-COMPLETE universe from profile 2026-09-06T215959Z -- scanning 20 of 394 screened coins (600 on disk;
stablecoins dropped).

['BTC/USDT',
'ETH/USDT',
'SOL/USDT',
'RE/USDT',
'XRP/USDT',
'BNB/USDT',
'PEPE/USDT',
'DOGE/USDT',
'SUI/USDT',
'WIF/USDT',
'AVAX/USDT',
'ENA/USDT',
'LINK/USDT',
'ADA/USDT',
'SHIB/USDT',
'TRX/USDT',
'ARB/USDT',
'WLD/USDT',
'DOT/USDT',
'LTC/USDT']

skip RE/USDT: KeyError
scored 19 pairs, 1 buy, 14 sell

  Symbol  Close ST_buy ST_agree Buy Sell RSI MACD_Buy
0  BTC/USDT 64020.0100 False   1 False True 40.94  False
1  ETH/USDT 1728.1900 False   1 False True 41.98  False
2  SOL/USDT  71.9500 False   2 False False 47.27  False
3  XRP/USDT  0.4761 False   0 False True 41.67   True
4  BNB/USDT 590.1500 False   0 False True 42.66  False
5  PEPE/USDT  0.0000 False   0 False True 35.27  False
6  DOGE/USDT  0.0825 False   0 False True 32.58  False
7  SUI/USDT  0.7198 False   0 False True 36.92  False
```

1.3 Performance Metrics

MACD is not retired; it feeds the model in three forms, each made scale-invariant so it fits a pipeline that compares hundreds of coins.

As a feature, the builder computes the PPO histogram (`f_ta_pta_ppo_hist`), which the code labels a scale-free MACD. PPO is MACD divided by price, so a sixty-thousand-dollar coin and a fraction-of-a-cent coin are comparable, which raw MACD is not. It sits in the optional pandas-ta momentum block beside TRIX, the Vortex indicator, CMO, the Fisher transform, and the Chande Kroll Stop, the wider momentum family the model carries.

As a cross-sectional signal, MACD and PPO are among the strength keys `cross_sectional_4h` ranks coins by, so momentum enters the relative-strength test in 1.2 directly.

As a standalone rule, 03-inputs/walkforward.py trades a daily MACD signal-line cross-up with ATR exits and an optional MACD cross-down exit. Scored after fees out of sample its verdict is NO-GO, one more confirmation that a single momentum trigger does not beat the fee on its own, which is why the model uses MACD as one ranked feature among many rather than as a trigger.

The methodology itself is unchanged: MACD reads momentum as the velocity of a trend, the 12-period EMA against the 26, with a crossover above the zero line bullish and below bearish and the histogram measuring how fast the lines converge or diverge. The board in 1.2 still reports a guarded crossover, MACD_Buy and MACD_Sell filtered by MACD_below_zero, as context beside the Supertrend baseline. The divergence taxonomy from the original chapter remains a design note, not implemented in the model.

1.4 Entry Points

The trend signals the model actually reads are the Supertrend family, not the illustrative board flags. The triple-Supertrend (`f_st_`) votes three ATR bands and the EMA-200 gate into an agreement score, signed distances, and a reversal flip. The Modern Adaptive Supertrend (`f_mst_`) adds a Kaufman efficiency-ratio regime gate and a commit filter that cuts false flips by around 80 percent. EMA-200 is the trend gate the live bot trails. The first signal not derived from the coin's own price is BTC lead-lag relative strength (`f_btc_`): BTC's own momentum, the coin's momentum relative to BTC, and a rolling beta and correlation, the market beta of crypto.

Part 3 finds that the entry rule alone, these trend signals used as a long trigger, sits at the efficient-market floor on the 1h frame: the trend reverses through the entry nearly as often as it continues. That is why the search moved to cross-sectional ranking and a coarser frame rather than to more entry features. The board's top candidates below illustrate an entry read on daily candles; they are not the decision-grade entry.

```
top = (results.sort_values(['ST_buy', 'Buy', 'EMA_gt_Kalman', 'AMAT', 'RSI'],
    ascending=[False, False, False, False, True])
    .reset_index(drop=True)).head(10))
bottom = (results.sort_values(['Sell', 'AMAT', 'RSI'], ascending=[False, False, False])
    .reset_index(drop=True)).head(10))
top[BOARD_COLS].style.map(color_boolean, subset=["ST_buy", "Buy", "Sell", "MACD_Buy"])
```

	Symbol	Close	ST_buy	ST_agree	Buy	Sell	RSI	MACD_Buy
0	WLD/USDT	0.619300	True	3	True	False	62.940000	False
1	LTC/USDT	44.560000	False	1	False	False	40.570000	False
2	SOL/USDT	71.950000	False	2	False	False	47.270000	False
3	ENA/USDT	0.090900	False	1	False	False	50.220000	False
4	TRX/USDT	0.334100	False	2	False	False	54.780000	False
5	ADA/USDT	0.158500	False	0	False	True	31.070000	False
6	DOGE/USDT	0.082500	False	0	False	True	32.580000	False
7	SHIB/USDT	0.000000	False	1	False	True	32.620000	False
8	DOT/USDT	0.937000	False	0	False	True	32.650000	False

	Symbol	Close	ST_buy	ST_agree	Buy	Sell	RSI	MACD_Buy
9	AVAX/USDT	6.241000	False	0	False	True	32.930000	False

<IPython.core.display.HTML object>

saved 04-outputs/HTML/chart.html

1.5 Exit Points

Exits are ATR-scaled, not signal-based. The stop and trailing stop are sized to each coin's volatility per frame and settled by the exit-geometry sweep, which fits per-coin trailing stops and a time-decaying take-profit and scores them after fees, as logged in 0.5. A MACD cross-down was tried as an optional signal exit in `walkforward.py` and left off by default, so the exit design is volatility geometry, not a momentum trigger.

Read exits before entries. The board's weakest-ranked names below are the illustrative exit read, flagged when none of the long conditions hold and price sits below the Kalman mean.

Exit candidates, downtrending with price below the Kalman estimate:

WIF/USDT
BNB/USDT
ETH/USDT
XRP/USDT
BTC/USDT
LINK/USDT
SUI/USDT
ARB/USDT
PEPE/USDT
AVAX/USDT

	Symbol	Close	ST_buy	ST_agree	Buy	Sell	RSI	MACD_Buy
0	WIF/USDT	0.162900	False	1	False	True	44.010000	False
1	BNB/USDT	590.150000	False	0	False	True	42.660000	False
2	ETH/USDT	1728.190000	False	1	False	True	41.980000	False
3	XRP/USDT	0.476100	False	0	False	True	41.670000	True
4	BTC/USDT	64020.010000	False	1	False	True	40.940000	False
5	LINK/USDT	7.874000	False	1	False	True	40.320000	False
6	SUI/USDT	0.719800	False	0	False	True	36.920000	False
7	ARB/USDT	0.083200	False	1	False	True	35.630000	False
8	PEPE/USDT	0.000000	False	0	False	True	35.270000	False
9	AVAX/USDT	6.241000	False	0	False	True	32.930000	False

The exit-geometry view is shared verbatim with Part 3 through `03-inputs/exit_geometry_viz.py`, so the exit points and the trend around each entry sit beside the ranking tables and cannot drift from the model's picture. It draws the three Supertrend trailing lines and EMA-200 over the survivorship 4h archive, shades the window before each entry, annotates the entry with its Supertrend agreement, EMA-200 side, and RSI, and tracks agreement, RSI, and volume beneath. These are the ATR-scaled

exits of 1.5 seen on the candles; their after-fee scoring lives in Part 3. This view is fixed to one representative resolution, BTC/USDT on the 4h frame ($frame=4$, the working frame), to illustrate the exit mechanics; it does not loop over 5m, 1h, and 4h. The multi-frame sweep is only in the Part 3 modelling sections.

[single] BTC/USDT 4h: 282 trades exits: 63 TP / 214 stop / 5 time mean net +0.07%/trade



[compare] BTC/USDT 4h: anchored on entry at bar 19113



1.6 Decision Checklist

The decision is not a board reading; it is the after-fee scoreboard. A signal counts only if it beats a coin flip, buy-and-hold, and the fee on a blind final year, and Part 3 runs that test per frame across 1h, 4h, and 5m. On the 1h frame the verdict is NO-GO, at the efficient-market floor; 4h carries a little more signal; the cross-sectional relative strength of 1.2 is the one real but still-underwater lead.

Two controls shape where a signal is allowed to trade. The regime state (f_{rg}) gives the model observable context: trailing realized volatility and its own-history percentile, a short and a long trend drift, a Kaufman trend efficiency, up-down breadth, and the BTC market regime. The regime gate in 0.5 then deploys the cross-sectional signal only when BTC is trending up, the cheapest in-mandate way to lift the underwater edge.

The daily board is demoted to what it is, an eyeball scan. Read exits before entries, treat a green cell as a candidate to check against the Part 3 evaluation, size and fence it in Part 2, and remember every snapshot is saved and dated under 04-outputs/. The signals table is written to CSV on each run.

saved ../04-outputs/CSV/DailySignals_20260908.csv

2. Trader Controls

The risk and rules layer, from Part 2 of this document. The boundaries every candidate strategy must live inside, above all the fee. This is a selective, high-conviction swing design, not a scalper: a trade held for a clean three to five percent move can absorb a 0.2 percent round trip, a trade chasing half a percent cannot, so the fee is the floor. Three guardrails enforce that, a cap on trades per day, a minimum net-edge floor on thin trades, and the after-fee validation protocol.

The control logic is demonstrated on synthetic OHLCV so it is legible without a data dependency; in production it consumes the same survivorship-complete archives as Parts 1 and 3. The 0.5 adjustments live here where they touch sizing and stops: the stop and trailing stop are ATR-scaled and provisional pending the exit-geometry sweep, and the regime gate is a new control layered on top and implemented in Part 3.

2.1 Key Variables

Every tunable parameter lives in one `CONFIG` block, never hard-coded, and the risk-bounding ones are operator-owned and held outside the model's reach: reviewed weekly, and when the bot is built, lifted into a locked module the strategy file cannot write to. Secrets stay in the operator's Keychain, never in the repo.

The block sets the venue fences (the round-trip fee and modelled slippage), the per-trade economics (the net-edge floor, the take-profit target, the ATR-based stop), the ATR band (floor and ceiling), and the hard selection gates. The stop and take-profit here are a starting point, provisional pending the exit-geometry sweep in Part 3 and superseded by the ATR-scaled exits logged in 0.5. A Supertrend band is itself an ATR-channel trailing stop, so it belongs to this controls vocabulary alongside the volatility filter and the sizing rule.

```
Environment ready · python 3.12.13  
ccxt 4.5.59 · pandas 2.3.3 · numpy 2.2.6 · plotly 6.8.0
```

```
Configured binance | account $750 | edge floor 1.5% | ATR band 2.5–12.0% | scan top 25
```

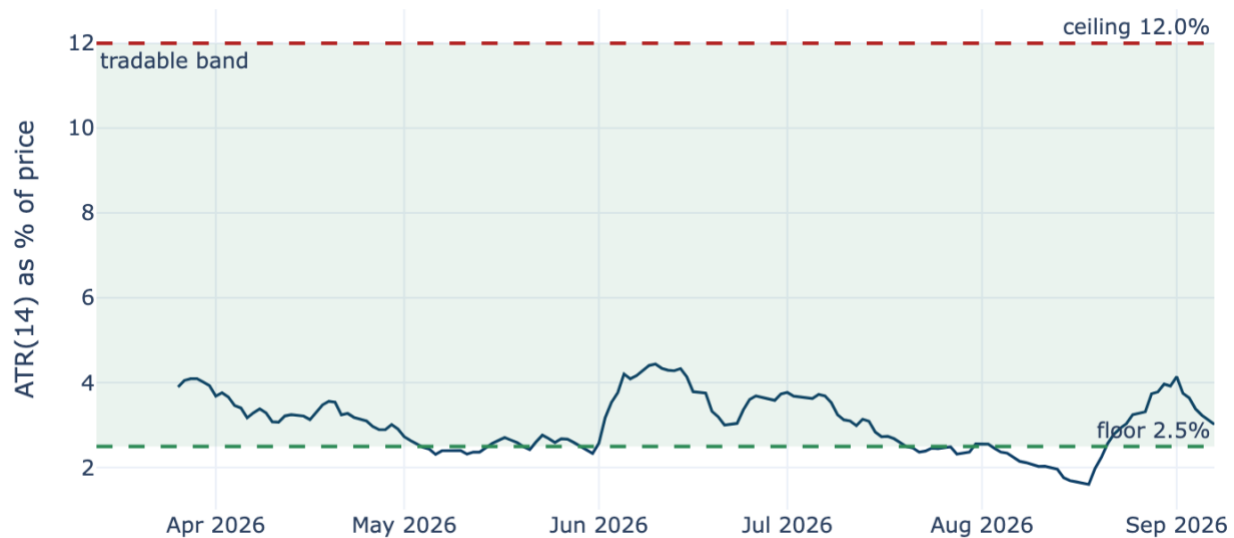
2.2 Volatility Filter

The control demos run on synthetic OHLCV so the logic is the point, not the bars; `synthetic_ohlc` and `fetch_daily` provide deterministic per-coin data offline, and the live path falls back to it if the network is unavailable. In production the same functions read the survivorship archives proven in Part 1.

The ATR band is the average true range over 14 days as a percent of price, a read of a coin's typical daily move, and it does two unrelated jobs. As a selection filter it admits or rejects a coin before the model sees it: below the floor, roughly two to three percent daily, a coin cannot reach the take-profit inside the hold window; above the ceiling it gaps through the stop and take-profit unpredictably and the indicators lose meaning. As a live guardrail it keeps the model from trading a coin that has drifted out of the tradable band.

```
universe (25): BTC/USDT, ETH/USDT, ZEC/USDT, SOL/USDT, XRP/USDT, BNB/USDT, DOGE/USDT, SUI/USDT, SOPH/USDT, WLD/USDT,  
UNI/USDT, ZKC/USDT ...
```


BTC/USDT — daily ATR% vs tradable band (selection + live guardrail)



2.3 Four-Gate Screen

Coins are screened individually each week on daily OHLCV plus the order-book top, through four gates. Gate 1 liquidity rejects anything below a 24-hour quote-volume floor, evaluated first because it is the cheap hard gate. Gate 2 is the ATR band, floor to ceiling as above. Gate 3 spread rejects anything wider than a tight ceiling well under the fee floor. Gate 4 history rejects tokens too young to carry a full indicator history. A coin passes only if all four hold, and the screen targets a shortlist of fifteen to twenty-five coins to hold positions on three or four.

The same four-gate screen is replayed point-in-time over history in Part 3, so each training row is kept only if it would have passed as of that bar. Because the archives carry no top-of-book spread, the historical replay approximates it with a Corwin-Schultz high-low estimator while the live screen uses the real spread. The candidate scatter places every scanned coin on liquidity against volatility, with survivors inside the band.

```
screen = run_screen(universe)
cols = ["symbol", "quote_volume_24h_usdt", "atr_pct", "spread_pct", "candle_count", "pass", "fail_reason"]
_view = screen[cols]
print(f"four-gate screen: {int(screen['pass'].sum())} of {len(screen)} pairs passed")
display(_view)
```

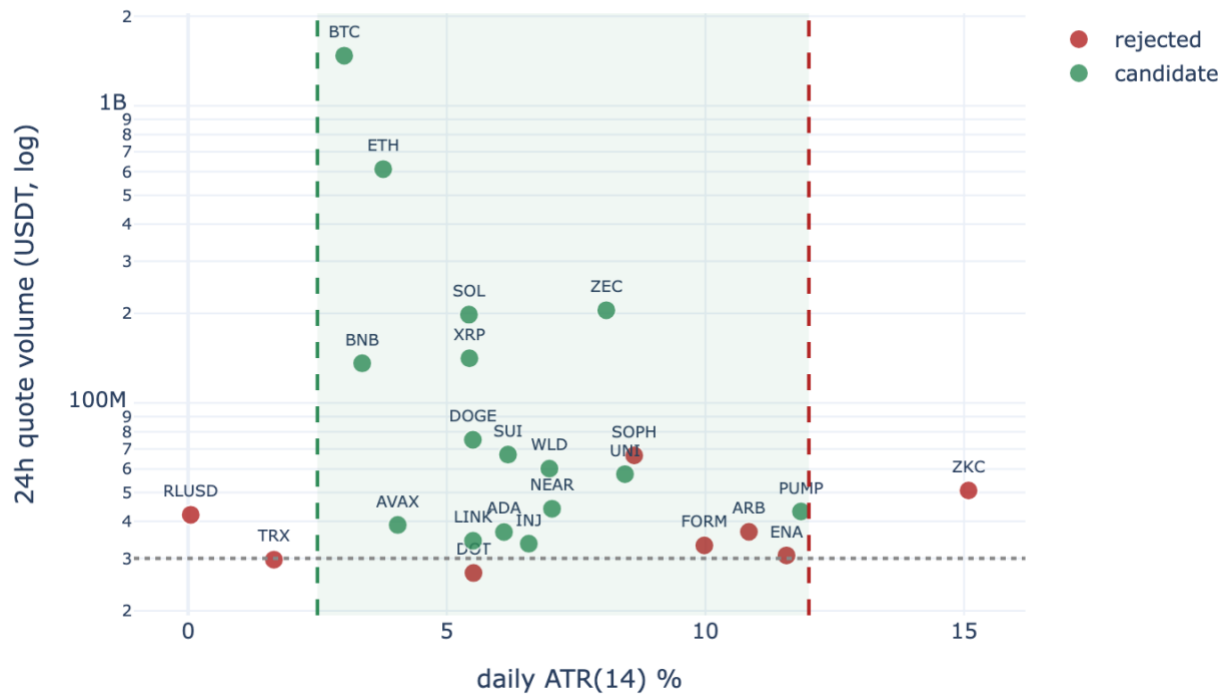
saved /Volumes/PortableSSD/Github/day-trader/04-outputs/Candidates_20260908.csv (16 of 25 passed)
four-gate screen: 16 of 25 pairs passed

	symbol	quote_volume_24h_usdt	atr_pct	spread_pct	candle_count \
0	BTC/USDT	1.473364e+09	3.017	0.0000	179
1	ETH/USDT	6.122891e+08	3.772	0.0004	179
2	ZEC/USDT	2.051502e+08	8.084	0.0009	179
3	SOL/USDT	1.983513e+08	5.429	0.0097	179
4	XRP/USDT	1.413853e+08	5.438	0.0071	179
5	BNB/USDT	1.360302e+08	3.365	0.0013	179
6	DOGE/USDT	7.521861e+07	5.505	0.0111	179
7	SUI/USDT	6.706095e+07	6.184	0.0121	179
8	WLD/USDT	6.018755e+07	6.984	0.0204	179
9	UNI/USDT	5.766502e+07	8.445	0.0146	179
10	NEAR/USDT	4.410371e+07	7.035	0.0415	179

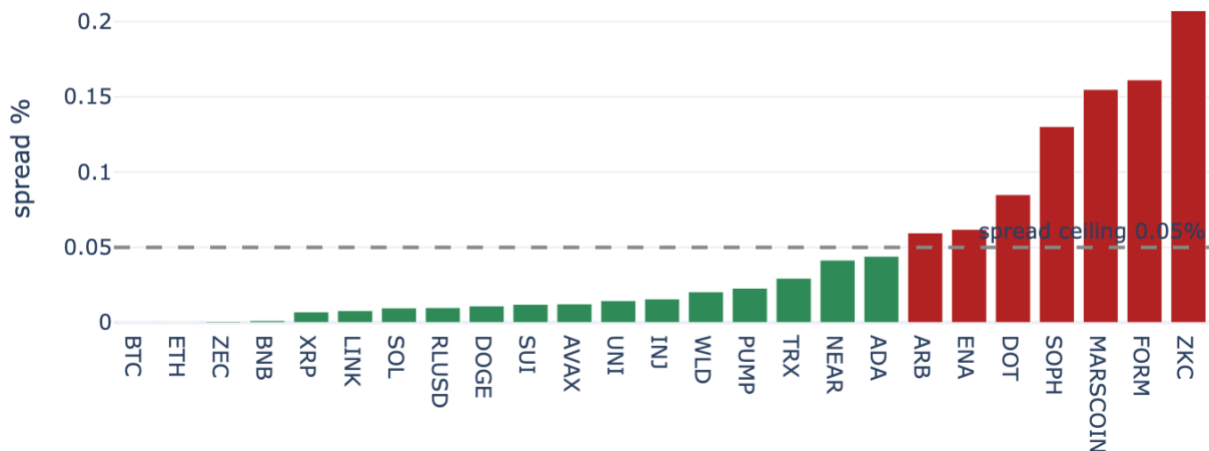
11	PUMP/USDT	4.315960e+07	11.847	0.0229	179
12	AVAX/USDT	3.888535e+07	4.052	0.0124	179
13	ADA/USDT	3.682421e+07	6.106	0.0440	179
14	LINK/USDT	3.440305e+07	5.508	0.0079	179
15	INJ/USDT	3.362874e+07	6.586	0.0157	179
16	SOPH/USDT	6.663541e+07	8.623	0.1303	179
17	ZKC/USDT	5.077757e+07	15.086	0.2073	179
18	MARSCOIN/USDT	4.775808e+07	NaN	0.1549	4
19	RLUSD/USDT	4.208149e+07	0.051	0.0100	179
20	ARB/USDT	3.688322e+07	10.841	0.0596	179
21	FORM/USDT	3.319087e+07	9.979	0.1613	179
22	ENA/USDT	3.071631e+07	11.567	0.0619	179
23	TRX/USDT	2.972413e+07	1.659	0.0295	179
24	DOT/USDT	2.681809e+07	5.517	0.0850	179

	pass	fail_reason
0	True	
1	True	
2	True	
3	True	
4	True	
5	True	
6	True	
7	True	
8	True	
9	True	
10	True	
11	True	
12	True	
13	True	
14	True	
15	True	
16	False	spread
17	False	atr_band,spread
18	False	atr_band,spread,history
19	False	atr_band
20	False	spread
21	False	spread
22	False	spread
23	False	liquidity,atr_band
24	False	liquidity,spread

Selection screen — liquidity vs volatility (survivors inside the band, above the



Spread gate — top-of-book spread vs ceiling



2.4 Edge Sizing

The net-edge fence is the entry gate: a trade is refused unless the estimated move minus the round-trip fee and slippage clears the edge floor. It is measured on net, not gross, and it refuses rather than warns. Exit coupling is enforced in code: the edge floor must sit below the take-profit, and the ATR floor must be high enough that a normal multi-day range reaches the take-profit inside the hold window.

Position sizing is volatility-scaled to keep dollar risk roughly constant across coins: the stop distance times the position value is held near a fixed risk budget, so a higher-ATR coin earns a smaller clip and a calmer coin a larger one, never past the per-position cap or below the venue minimum. At a small account this pushes toward a few larger positions rather than many tiny clips that waste edge on fees. The take-profit and stop in the CONFIG reference an older fixed label and are provisional pending the exit-geometry and label sweeps in 0.5 and Part 3.

```
print("Net-edge fence, gross estimated move to net, and admission:")
for m in (1.0, 1.5, 2.0, 3.5):
    print(f"est {m}>4}% -> net {net_edge(m)>5.2f}% -> {'ALLOW' if net_edge_ok(m) else 'refuse'}")

print("\nExit-coupling checks (config must satisfy all):")
for k, v in validate_exit_coupling().items():
    print(f" [{k} {'if v else 'XX'}] {k}")
```

Net-edge fence, gross estimated move to net, and admission:

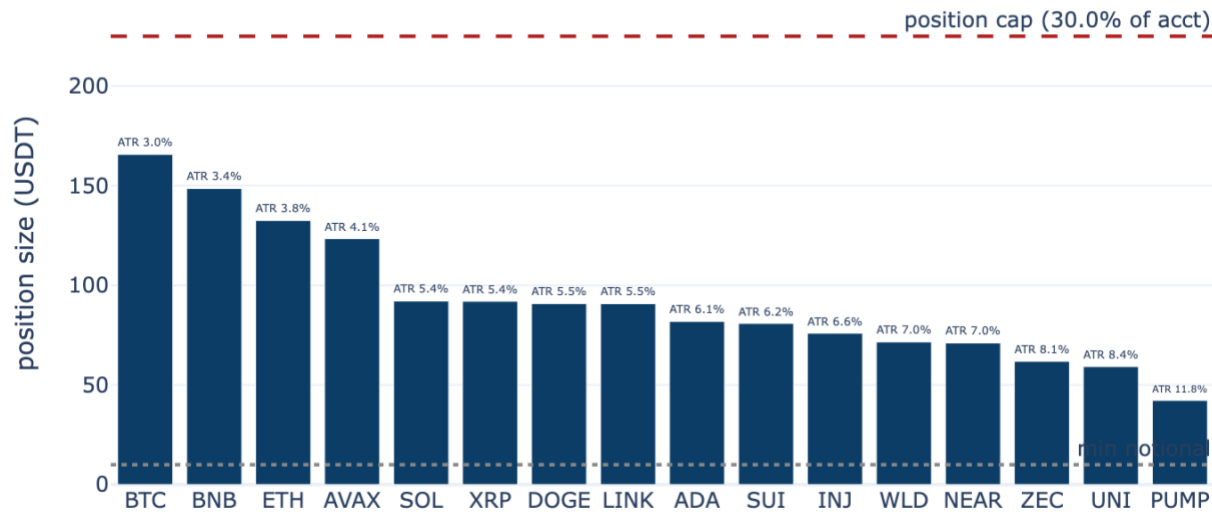
```
est 1.0% -> net 0.80% -> refuse
est 1.5% -> net 1.30% -> refuse
est 2.0% -> net 1.80% -> ALLOW
est 3.5% -> net 3.30% -> ALLOW
```

Exit-coupling checks (config must satisfy all):

```
[ok] edge_floor < take_profit
[ok] round_trip < edge_floor
[ok] take_profit > 2x round_trip (alive)
[ok] TP reachable at the ATR floor within the window (about 4.0 days)
```

	symbol	atr_pct	stop_pct	position_usdt	floored_at_min
0	BTC/USDT	3.017	4.526	165.73	False
1	BNB/USDT	3.365	5.048	148.59	False
2	ETH/USDT	3.772	5.658	132.56	False
3	AVAX/USDT	4.052	6.078	123.40	False
4	SOL/USDT	5.429	8.143	92.10	False
5	XRP/USDT	5.438	8.157	91.95	False
6	DOGE/USDT	5.505	8.258	90.83	False
7	LINK/USDT	5.508	8.262	90.78	False
8	ADA/USDT	6.106	9.159	81.89	False
9	SUI/USDT	6.184	9.276	80.85	False
10	INJ/USDT	6.586	9.879	75.92	False
11	WLD/USDT	6.984	10.476	71.59	False
12	NEAR/USDT	7.035	10.553	71.07	False
13	ZEC/USDT	8.084	12.126	61.85	False
14	UNI/USDT	8.445	12.668	59.21	False
15	PUMP/USDT	11.847	17.770	42.20	False

ATR-scaled position sizing: constant \$ risk (higher ATR means a smaller clip)



2.5 Hold Period

The hold period is the swing band, days to about two weeks. Scalping and day-trading are ruled out by fees and the pattern-day-trading rule, position-trading by underusing the signal stack. Within the swing band the hold window is a walk-forward parameter, searched roughly one to twenty days and picked by out-of-sample results, not by a gut figure. Shorter holds demand a higher ATR to reach the target in time; longer holds tolerate a lower one. This is the same coupling as the edge fence, seen on the time axis.

2.6 Stacked Defences

The defences are layered, not single. The selection screen keeps unsuitable coins out, the net-edge fence keeps thin trades out, the ATR band is both a selection filter and a live guardrail, the position cap and venue minimum bound each clip, and the per-day trade cap and the cash floor bound the book. Above them sit the 0.5 controls: the ATR-scaled stop and trailing stop, provisional pending the exit-geometry sweep, and the new regime gate that only lets the cross-sectional signal trade when BTC is trending up, implemented in Part 3.

The floor fence is the minimum net edge; the venue fences are the fee and slippage assumptions that define it. Each defence is cheap and independent, so a coin must clear all of them and any one can veto.

2.7 Signal Journal

A learning aid, not a trader. For each coin it finds every entry and exit the MACD logic produced, draws the candles with the moving averages and a volume histogram, marks each signal, and writes a table beside the chart recording what the model saw, trend, RSI, and volatility, and whether the fee fence would actually let the trade fire. Each study sheet is saved as a self-contained HTML page under `04-outputs/journal/`, so a visual library builds up on every run. The net-edge column uses the take-profit as a stand-in for the expected move until the Part 3 model supplies a

real per-trade estimate; the entries marked would-fire are the ones that clear both the fee floor and the ATR band.

saved /Volumes/PortableSSD/Github/day-trader/04-outputs/journal/journal_BTC-USDT_20260908.html - 22 entries/exits documented
 saved /Volumes/PortableSSD/Github/day-trader/04-outputs/journal/journal_ETH-USDT_20260908.html - 14 entries/exits documented
 saved /Volumes/PortableSSD/Github/day-trader/04-outputs/journal/journal_ZEC-USDT_20260908.html - 18 entries/exits documented

3 journal(s) saved under 04-outputs/journal/

--- BTC/USDT ---



	date	action	price	trend	rsi	atr_pct	net_edge_pct \
0	2026-02-02	enter / buy	78738.61	up	NaN	NaN	9.8
1	2026-02-03	exit / sell	75770.21	up	NaN	NaN	9.8
2	2026-02-20	enter / buy	68020.01	down	41.0	4.27	9.8
3	2026-02-23	exit / sell	64656.02	down	33.0	3.95	9.8
4	2026-02-25	enter / buy	67988.04	down	53.0	4.00	9.8
5	2026-03-21	exit / sell	68918.12	up	55.0	3.86	9.8

```

6 2026-04-06 enter / buy 68853.66 down 42.0 3.17 9.8
7 2026-04-28 exit / sell 76342.77 up 57.0 2.89 9.8
8 2026-05-04 enter / buy 79861.01 up 67.0 2.48 9.8
9 2026-05-12 exit / sell 80504.47 up 68.0 2.36 9.8
10 2026-06-13 enter / buy 64458.01 down 27.0 4.28 9.8
11 2026-06-27 exit / sell 60029.00 down 33.0 3.66 9.8
12 2026-06-29 enter / buy 60260.21 down 25.0 3.58 9.8
13 2026-06-30 exit / sell 58624.71 down 23.0 3.74 9.8
14 2026-07-01 enter / buy 60024.00 down 33.0 3.77 9.8
15 2026-07-27 exit / sell 63755.86 up 56.0 2.49 9.8
16 2026-08-07 enter / buy 64923.19 up 54.0 2.15 9.8
17 2026-08-11 exit / sell 63600.00 up 48.0 2.03 9.8
18 2026-08-18 enter / buy 64725.42 down 56.0 1.60 9.8
19 2026-09-02 exit / sell 77340.01 up 70.0 3.75 9.8
20 2026-09-03 enter / buy 81270.37 up 71.0 3.64 9.8
21 2026-09-04 exit / sell 79660.77 up 54.0 3.38 9.8

```

```

fence size_usdt \
0 blocked: out of band NaN
1 exit NaN
2 would fire 117.03
3 exit NaN
4 would fire 124.88
5 exit NaN
6 would fire 157.60
7 exit NaN
8 blocked: out of band NaN
9 exit NaN
10 would fire 116.78
11 exit NaN
12 would fire 139.55
13 exit NaN
14 would fire 132.64
15 exit NaN
16 blocked: out of band NaN
17 exit NaN
18 blocked: out of band NaN
19 exit NaN
20 would fire 137.28
21 exit NaN

```

```

note
0 MACD crossed above signal -> momentum turning up
1 MACD crossed below signal -> momentum turning ...
2 MACD crossed above signal -> momentum turning up
3 MACD crossed below signal -> momentum turning ...
4 MACD crossed above signal -> momentum turning up
5 MACD crossed below signal -> momentum turning ...
6 MACD crossed above signal -> momentum turning up
7 MACD crossed below signal -> momentum turning ...
8 MACD crossed above signal -> momentum turning up
9 MACD crossed below signal -> momentum turning ...
10 MACD crossed above signal -> momentum turning up
11 MACD crossed below signal -> momentum turning ...
12 MACD crossed above signal -> momentum turning up
13 MACD crossed below signal -> momentum turning ...
14 MACD crossed above signal -> momentum turning up
15 MACD crossed below signal -> momentum turning ...
16 MACD crossed above signal -> momentum turning up
17 MACD crossed below signal -> momentum turning ...
18 MACD crossed above signal -> momentum turning up
19 MACD crossed below signal -> momentum turning ...
20 MACD crossed above signal -> momentum turning up
21 MACD crossed below signal -> momentum turning ...

```


2.8 Frozen Harness

The yardstick is frozen and operator-owned. Every walk-forward run appends one line to a log, the parameters tried, the out-of-sample after-fee result, and whether it was kept or discarded, and the operator reviews it weekly. The model never writes to the harness or the log, which keeps the evaluation surface human rather than automated away. The harness fixes the out-of-sample window, the train window, the embargo that straddles the cut, the regime set, the fee assumption, and the metric, identical across every comparison so results are commensurable.

The dated candidate table and the experiment log are written under 04-outputs/ each run, and nothing else: the notebook screens and sizes only, no orders, with LIVE_TRADING left false and never read here.

logged -> /Volumes/PortableSSD/Github/day-trader/04-outputs/experiment_log.csv
experiment_log.csv, 28 rows, last 3:

```

      ts_utc  harness  kept \
25 2026-09-08T14:06:26+00:00 harness-v1 False
26 2026-09-08T15:22:52+00:00 harness-v1 False
27 2026-09-08T15:35:28+00:00 harness-v1 False

      why  p_edge_floor_pct \
25 no demonstrated OOS edge after fees (matches s...      1.5
26 no demonstrated OOS edge after fees (matches s...      1.5
27 no demonstrated OOS edge after fees (matches s...      1.5

p_take_profit_pct p_atr_floor_pct p_threshold \
25      10.0      2.5      2
26      10.0      2.5      2
27      10.0      2.5      2

oos_after_fee_expectancy_pct oos_win_rate oos_n_trades \
25      -0.4      0.41      37
26      -0.4      0.41      37
27      -0.4      0.41      37

oos_vs_buy_hold_pct
25      1.2
26      1.2
27      1.2

```

Artefacts written this run:

```

/Volumes/PortableSSD/Github/day-trader/04-outputs/Candidates_20260629.csv (1946 bytes)
/Volumes/PortableSSD/Github/day-trader/04-outputs/Candidates_20260701.csv (1956 bytes)
/Volumes/PortableSSD/Github/day-trader/04-outputs/Candidates_20260703.csv (1995 bytes)
/Volumes/PortableSSD/Github/day-trader/04-outputs/Candidates_20260906.csv (1858 bytes)
/Volumes/PortableSSD/Github/day-trader/04-outputs/Candidates_20260907.csv (1845 bytes)
/Volumes/PortableSSD/Github/day-trader/04-outputs/Candidates_20260908.csv (1833 bytes)
/Volumes/PortableSSD/Github/day-trader/04-outputs/experiment_log.csv (3970 bytes)

```

Reminder: this notebook screens and sizes only. No orders are placed; LIVE_TRADING stays 'false' in 03-inputs/config.py and is never read here.

3. Trader Execution

The modeling and search layer, from Part 3 of this document. Where the datasets, features, and model meet the after-fee test, run per frame across 1h, 4h, and 5m. This is the decision-grade chapter: everything Parts 1 and 2 describe is built, screened point-in-time, labelled, trained, and scored on a blind final year here, and only what clears the fee out of sample counts.

The 0.5 adjustments are implemented across these sections: the label in 3.1 and 3.7, the ATR-scaled exits in 3.3, the regime gate in 3.14, and the longer-horizon frame across every per-frame loop. The cells run top to bottom on the project .venv. LOAD_FRAME selects the active frame the single-frame cells consume; the multi-frame sections from 3.9 on loop over all three frames on their own and restore the active frame afterward.

3.1 Data Preparation

The environment imports the shared loader `build_dataset_1h`, the split and cost machinery, and the model libraries. The three frames are built offline from the survivorship-complete `data.binance.vision` archives into `dataset_5m_allmarket`, `dataset_1h_allmarket`, and `dataset_4h_allmarket`, one builder retuned per frame. `LOAD_FRAME` picks the active frame, Import Data loads its Parquet into `df` and `feat`, and the train-test split holds out the final year with an embargo equal to the label horizon so no future leaks across the cut. The label is the ATR-scaled triple barrier; the 0.5 revision toward a longer, less fee-punishing geometry is the intended change tracked here.

```
outputs -> /Volumes/PortableSSD/Github/day-trader/04-outputs
env ready . kernel: /Volumes/PortableSSD/Github/day-trader/.venv/bin/python
python 3.12.13 . lightgbm yes . pandas-ta yes . TA-Lib yes

label: +2.0 ATR before -1.0 ATR within 12 bars (2d) . hold out final 365d, embargo +/-2d . confidence 0.4-0.6 . cost 0.20% round trip

SOURCE / LOCATION (Binance spot, data.binance.vision archives; one dataset per frame)
[ 1h] dataset_1h_allmarket.parquet 2026-06-21 11:15, 684 MB <- active
      klines: /Volumes/PortableSSD/Github/day-trader/03-inputs/binance-data/klines_1h
[ 4h] dataset_4h_allmarket.parquet 2026-06-23 20:30, 2008 MB
      klines: /Volumes/PortableSSD/Github/day-trader/03-inputs/binance-data/klines_4h
[ 5m] dataset_5m_allmarket.parquet 2026-06-24 23:02, 2990 MB
      klines: /Volumes/PortableSSD/Github/day-trader/03-inputs/binance-data/klines_5m
storage : Parquet (columnar, compressed, dtype-preserving); CSV is the fallback

ACTIVE FRAME = 1h (the cells below model this frame)
flow table : flow_1h.parquet 2026-06-21 10:36, 458 MB
dataset    : dataset_1h_allmarket.parquet 2026-06-21 11:15, 684 MB

TIME WINDOWS (counts of bars; 1 bar = the active frame)
wall-clock (daily x bars/day): {'ema_fast': 336, 'ema_mid': 2184, 'ema_slow': 3000, 'rsi': 336, 'bb': 336, 'bb_std': 2.0, 'atr': 336, 'rv_short': 168, 'rv_long': 720, 'vol': 480, 'mom': [120, 240, 480, 1440]}
intraday      : {'ema_fast': 12, 'ema_mid': 26, 'ema_slow': 50, 'rsi': 14, 'bb': 20, 'bb_std': 2.0, 'atr': 14, 'rv_short': 24, 'rv_long': 168, 'vol': 24, 'mom': [6, 12, 24, 72, 168]}

LABEL (ATR-scaled triple barrier): {'tgt_atr': 2.0, 'stp_atr': 1.0, 'horizon_bars': 48, 'atr_len': 14}
SCREEN gates      : {'min_quote_volume_usdt': 300000000, 'qv_window': 24, 'atr_floor_pct': 0.5, 'atr_ceiling_pct': 2.5, 'atr_len': 14, 'min_history_bars': 2880, 'spread_proxy_ceiling_pct': 0.5, 'spread_window': 24}
DATA-QUALITY gate : {'max_gap_ratio': 0.02, 'max_single_gap_hours': 72}

To (re)build offline from the .venv (reference - not run here):
.venv/bin/python 03-inputs/binance-data/flow_data.py --interval 1h --all-market
.venv/bin/python 03-inputs/build_dataset_1h.py --interval 1h

LOADED 1,640,660 rows from Parquet (1h frame)
in-sample rows : 458,539 (27.9% of total)
coins          : 47
features       : 61
date range     : 2017-10-16 11:00:00 -> 2026-06-18 23:00:00
base rate all / in-sample : 0.322 / 0.313
feature families : wall-clock 13, intraday 14, in-house TA 8, pandas-ta 7, TA-Lib 15, flow 4
in-sample rows with any NaN feature: 0
rows per coin:
```

	rows
symbol	
BTC/USDT	75229
ETH/USDT	75229
BNB/USDT	73293
LTC/USDT	72405
ADA/USDT	69437
LINK/USDT	62715
FET/USDT	61837
DASH/USDT	61177
DOGE/USDT	58971
HBAR/USDT	56858
CHZ/USDT	56837
BCH/USDT	55495
SOL/USDT	49343
JST/USDT	49290
SAND/USDT	49264
PAXG/USDT	48754
BEL/USDT	48437
AVAX/USDT	48335
NEAR/USDT	47808
AAVE/USDT	47772
INJ/USDT	47613
AXS/USDT	47352
ALICE/USDT	44456
RIF/USDT	44034
BICO/USDT	37805

3.2 Feature Variables

Every feature is causal and scale-invariant, computed from past bars only and normalized so hundreds of coins compare. The families span an in-house wall-clock and intraday baseline, extra oscillators, an optional pandas-ta momentum block including the scale-free PPO, an optional TA-Lib block, the trade-flow imbalance, the triple-Supertrend (`f_st_`), the Modern Adaptive Supertrend (`f_mst_`), BTC lead-lag relative strength (`f_btc_`), multi-timeframe context (`f_4h_`, `f_d1_`, `f_w1_`), and the

regime state (f_{rg}). Feature integration verifies the higher-timeframe joins are causal with no lookahead.

```
wall-clock (wc) : 13
intraday (hr)  : 14
in-house TA    : 8
pandas-ta      : 7
TA-Lib         : 15
flow          : 4
TOTAL         : 61
```

	family	mean	std	min	max
0	f_flow	0.117	0.067	-0.466	0.668
1	f_hr	0.503	0.122	0.054	0.965
2	f_hr_a	0.013	0.005	0.005	0.025
3	f_hr_b	0.503	0.337	-0.576	1.582
4	f_hr_e	0.000	0.011	-0.070	0.054
5	f_hr_m	0.005	0.058	-0.331	0.502
6	f_hr_r	0.326	0.109	0.054	0.905
7	f_hr_v	1.024	0.789	0.002	19.739
8	f_ta	0.138	0.321	-1.395	1.737
9	f_ta_a	0.020	0.606	-1.000	1.000
10	f_ta_d	-0.007	0.148	-0.721	0.754
11	f_ta_p	-0.015	0.392	-1.647	1.529
12	f_tl	0.512	0.085	0.127	0.864
13	f_tl_c	0.017	0.156	-0.600	0.700
14	f_tl_h	0.648	0.524	-0.125	1.375
15	f_tl_m	0.000	0.013	-0.110	0.167
16	f_tl_s	0.000	0.033	-0.274	0.508
17	f_wc	0.502	0.026	0.372	0.647
18	f_wc_a	0.014	0.006	0.002	0.076
19	f_wc_b	0.531	0.358	-1.825	2.594
20	f_wc_e	-0.012	0.129	-1.147	0.404
21	f_wc_m	0.064	0.358	-0.746	10.776
22	f_wc_r	0.325	0.085	0.087	0.714
23	f_wc_v	1.032	1.123	0.001	129.313

	prefix	block	columns
0	f_st_	triple Supertrend, three ATR bands and the EMA-200 gate	0
1	f_mst_	adaptive Supertrend with a Kaufman efficiency gate	0
2	f_btc_	relative strength against the market factor	0
3	f_4h_	four-hour context, merged causally	0
4	f_d1_	daily context, merged causally	0
5	f_w1_	weekly context, merged causally	0
6	f_rg_	regime state: realised volatility, drift, breadth	0
7	f_flow_	taker buy and sell imbalance	4

DATASET DIMENSIONS & CHARACTERISTICS

	value
in-sample rows	458,539
coins	38
features	61
date span	2017-12-15 to 2026-06-18
base rate (P win)	0.313
label 1 / 0	143,615 / 314,924
rows with any NaN feature	0
in-memory size	150 MB

label base rate by year:

	rows	base_rate
year		
2017	114	0.307
2018	13390	0.293

	rows	base_rate
year		
2019	19570	0.306
2020	32475	0.334
2021	68261	0.318
2022	70764	0.308
2023	51624	0.321
2024	85228	0.312
2025	91006	0.314
2026	26107	0.290

top 10 coins carry 81% of rows (panel tilts to long-history coins):

	rows	row_share	base_rate	first	last
symbol					
ETH/USDT	59543	0.130	0.324	2017-12-18	2026-06-18
BTC/USDT	54154	0.118	0.313	2017-12-15	2026-06-18
BNB/USDT	47723	0.104	0.320	2018-03-19	2026-06-18
SOL/USDT	35692	0.078	0.318	2021-02-04	2026-06-18
DOGE/USDT	35421	0.077	0.316	2021-02-27	2026-06-18
ADA/USDT	33338	0.073	0.305	2018-09-23	2026-06-18
LTC/USDT	32527	0.071	0.307	2018-05-03	2026-06-06
LINK/USDT	29987	0.065	0.312	2019-10-09	2026-06-07
AVAX/USDT	23799	0.052	0.316	2021-02-21	2026-06-06
BCH/USDT	17200	0.038	0.315	2020-03-27	2026-06-04

most correlated feature pairs (redundancy):

	feature_a	feature_b	abs_corr
0	f_ta_willr	f_ta_stochk	1.000
1	f_wc_rv_short	f_hr_rv_long	1.000
2	f_hr_rsi	f_ta_pta_cmo	1.000
3	f_flow_taker_ratio	f_flow_imb	1.000
4	f_ta_cci	f_hr_bb_pos	0.940

	feature_a	feature_b	abs_corr
5	f_hr_ema_mid_slow	f_ta_pta_trix	0.935
6	f_hr_rsi	f_ta_dmi_diff	0.933
7	f_ta_pta_cmo	f_ta_dmi_diff	0.933
8	f_hr_ema_fast_mid	f_hr_mom_24	0.914
9	f_ta_pta_trix	f_hr_mom_72	0.894

top 12 features by class separation |mean(1) - mean(0)| / sd:

	separation
f_ta_pta_vortex_diff	0.059
f_ta_dmi_diff	0.059
f_ta_mfi	0.054
f_ta_pta_ppo_hist	0.054
f_ta_pta_cmo	0.050
f_hr_rsi	0.050
f_hr_mom_12	0.050
f_ta_pta_cksp_long_dist	0.044
f_ta_cci	0.044
f_hr_bb_pos	0.041
f_hr_ema_fast_mid	0.040
f_wc_bb_pos	0.038

3.3 Entries Exits

The entry and exit mechanics are drawn on real candles so the picture and the after-fee scoreboard cannot drift, all through the shared `exit_geometry_viz`. The panels show OHLCV with the three Supertrend trailing bands and EMA-200, the entry and exit points across regimes and multiple timelines, the entry design on candles, the exit geometry with its ratcheting trailing stop and decaying take-profit, and a side-by-side comparison of exit configurations. The exits are the ATR-scaled, per-frame geometry of 0.5, seen here before they are scored.



BTC/USDT: 1145 trades total . 5 entries / 6 exits in the 220-bar window



ETH/USDT: 1166 trades total . 6 entries / 6 exits in the 220-bar window



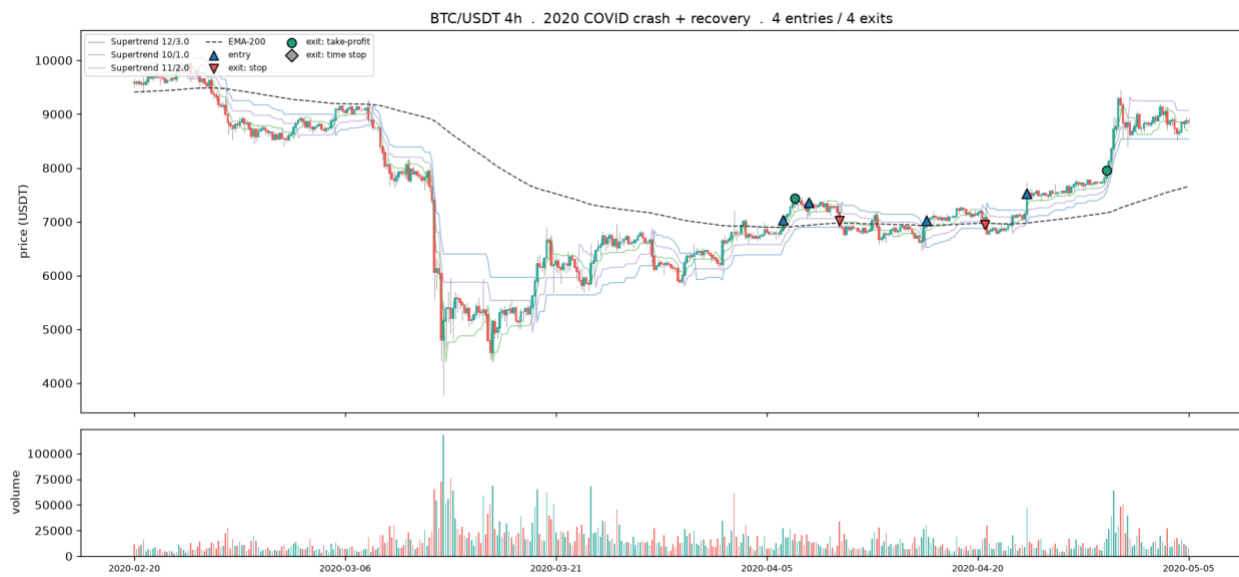
SOL/USDT: 820 trades total . 6 entries / 7 exits in the 220-bar window



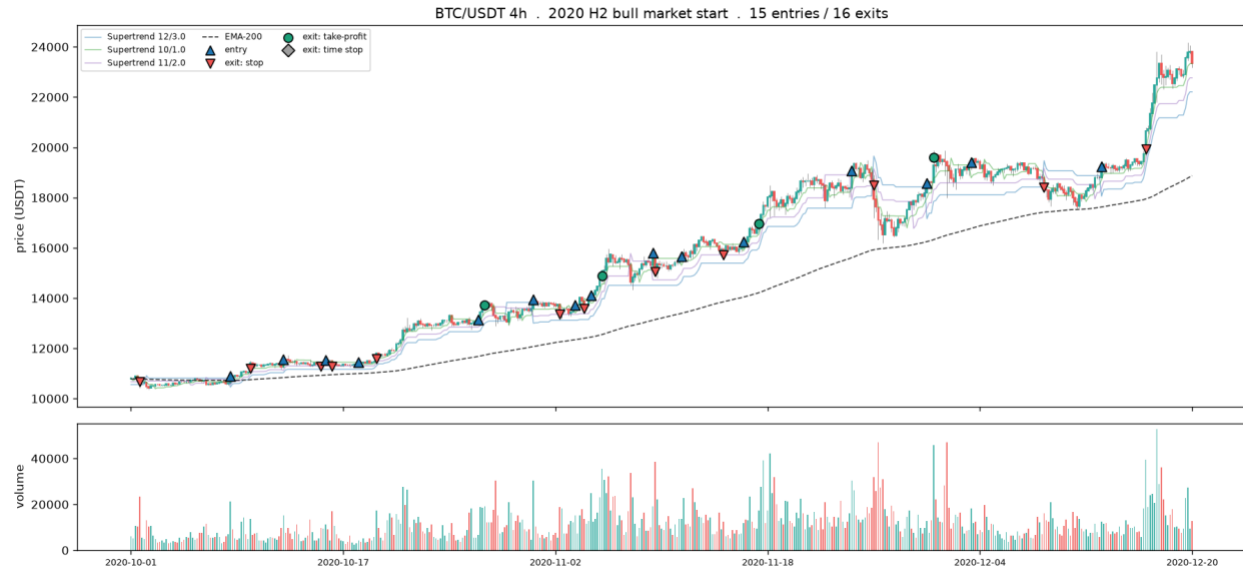
DOGE/USDT: 804 trades total . 2 entries / 2 exits in the 220-bar window



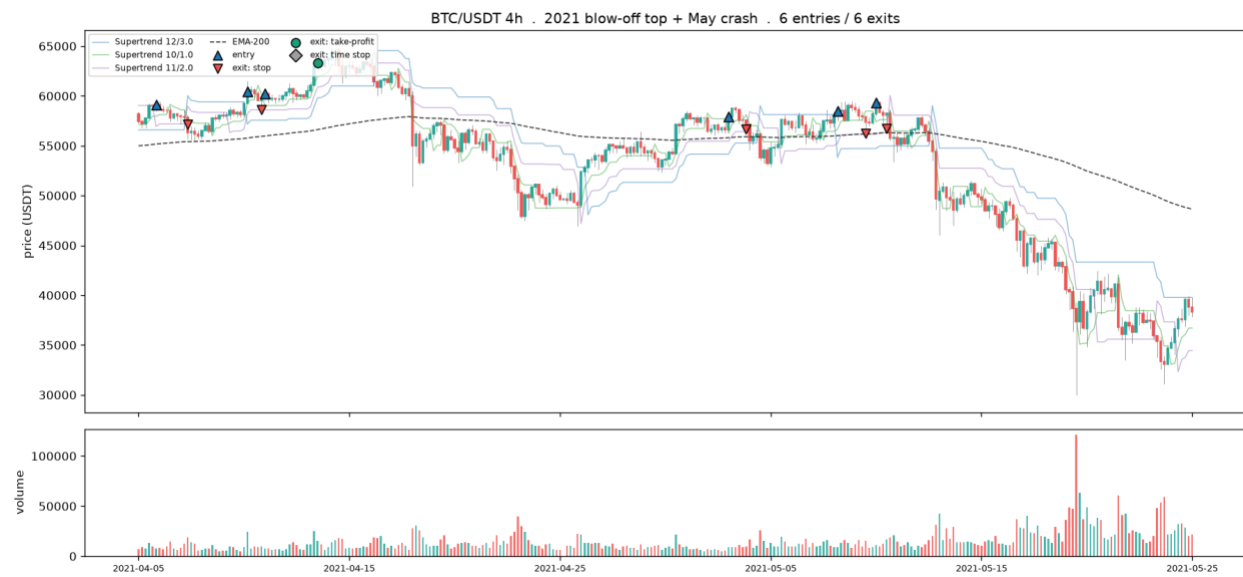
2019 spring rally 2019-04-01 -> 2019-06-20 13 entries / 13 exits



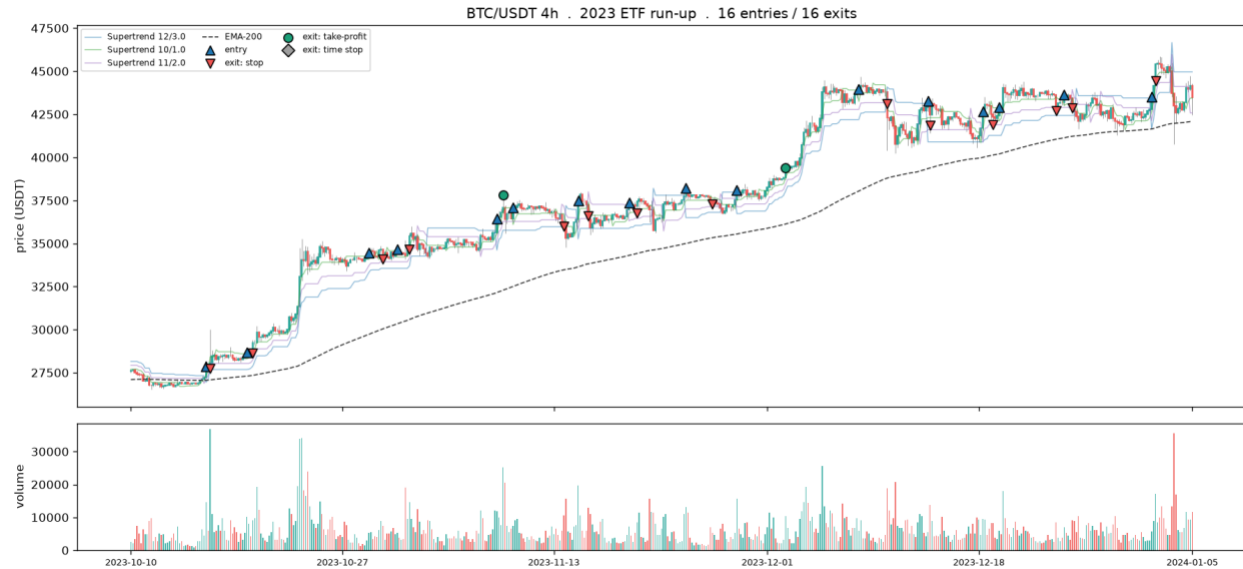
2020 COVID crash + recovery 2020-02-20 -> 2020-05-05 4 entries / 4 exits



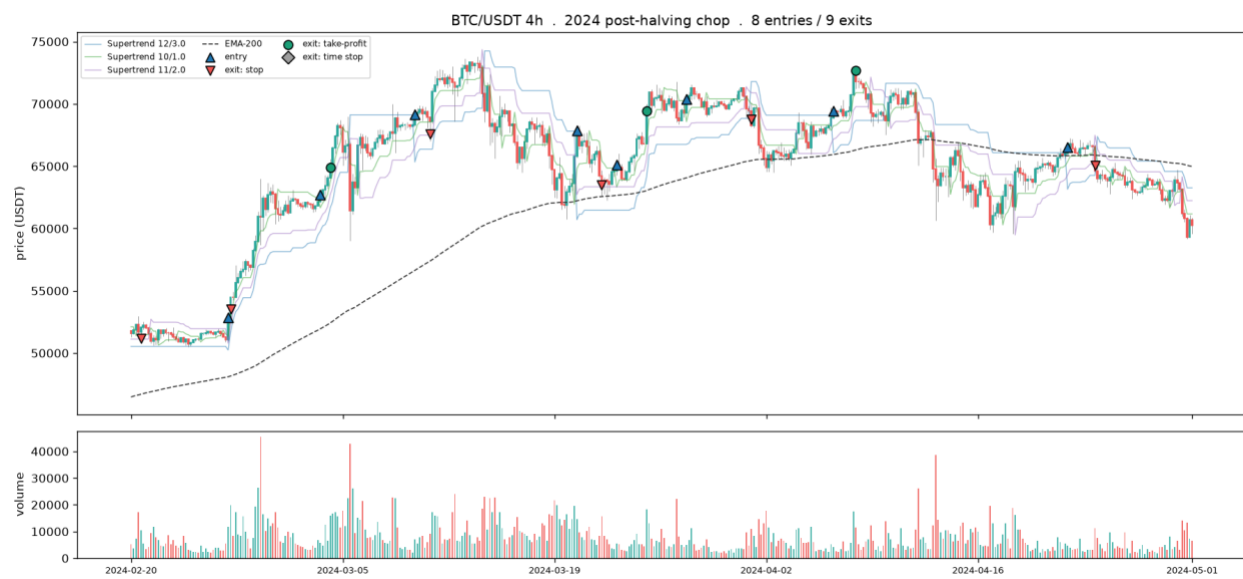
2020 H2 bull market start 2020-10-01 -> 2020-12-20 15 entries / 16 exits



2021 blow-off top + May crash 2021-04-05 -> 2021-05-25 6 entries / 6 exits



2023 ETF run-up 2023-10-10 -> 2024-01-05 16 entries / 16 exits



2024 post-halving chop 2024-02-20 -> 2024-05-01 8 entries / 9 exits

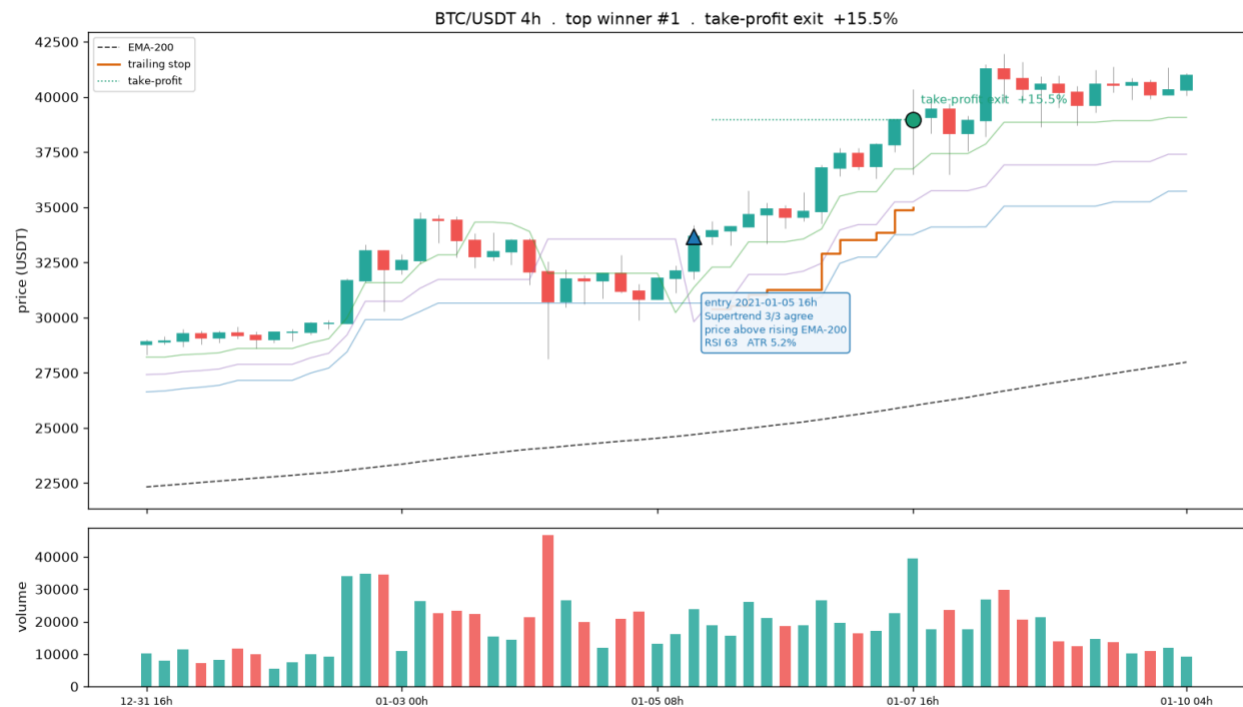
note: the 2022 LUNA/bear leg is omitted because the rule fires ZERO entries there -- it stands aside when no uptrend exists, which is itself the correct (and only safe) behaviour.





1h vs 4h at equal candle count is lever 1 (frame); 220 vs 700 on 4h is lever 2 (window). Past ~900 candles, step to a coarser frame rather than zoom further.

from 282 BTC 4h trades . best +15.5% / worst -8.7%







window 2026-06-12 08h -> 2026-06-18 23h | 111 labelled entries, 35 wins (32%), base rate 31%

[single] BTC/USDT 1h: 1145 trades exits: 181 TP / 963 stop / 1 time mean net -0.35%/trade -> /Volumes/PortableSSD/Github/day-trader/04-outputs/PNG/3-exit-geometry-candles.png



[compare] BTC/USDT 1h: anchored on entry at bar 77414 -> /Volumes/PortableSSD/Github/day-trader/04-outputs/PNG/3-exit-geometry-compare.png



3.4 Sharpening Entries

A walk-forward test of whether entry conditioning improves expectancy. Range and volatility conditions do not, trend and momentum do so marginally, and an early BTC-trend-gated entry that fires a few bars before the Supertrend flip lifts a single year to breakeven but does not survive walk-forward, positive in only a minority of folds. The conclusion carried through the rest of the chapter is that more price-derived entry features will not cross the fee line, which is why the search turns to cross-sectional framing and a coarser frame.

Highest-ranked single-feature entry conditions, in-sample and gross of cost:

	feature	best_decile_winrate	lift_over_breakeven
0	f_ta_pta_vortex_diff	0.342	0.009
1	f_hr_mom_12	0.335	0.001
2	f_ta_dmi_diff	0.333	-0.000
3	f_ta_mfi	0.333	-0.000
4	f_ta_pta_cmo	0.331	-0.003
5	f_hr_rsi	0.331	-0.003
6	f_ta_pta_ppo_hist	0.327	-0.006
7	f_ta_pta_cksp_long_dist	0.323	-0.010

cost drag is ~0.20% round trip; on a +2.0 ATR target a win must clear breakeven by more than that to pay.

3.5 Survivorship Pipeline

The three-stage pipeline that makes the panel survivorship-complete and point-in-time. Stage A crawls the `data.binance.vision` listing to enumerate every USDT pair ever listed, delisted coins included, and downloads them checksum-verified. Stage B profiles coverage, gaps, listing timelines, and liquidity to derive the usable start, the minimum history, and the purge and embargo. Stage C is the forward-chained walk-forward splitter with per-fold point-in-time universes and train-only transforms. This is the spine Parts 1 and 2 borrow their universe from.

```
historical USDT universe (incl. dead coins): 683 pairs (+52 leveraged excluded)
dead-coin audit: 10/10 known-delisted present in universe
snapshot: 487 active USDT pairs as of 2026-09-08 -> /Volumes/PortableSSD/Github/day-trader/03-inputs/binance-
data/exchange_info_snapshot_2026-09-08.json
survivorship: 683 universe / 480 active / 203 delisted = 29.7% dead coins

profile run : /Volumes/PortableSSD/Github/day-trader/04-outputs/3A-training-test-data/panel-profile/2026-09-06T215959Z
symbols    : 600 | delisted share: 33.3%
usable start : 2019-04-01
min history : 157 days
purge/embargo: 125 days (3000 bars)
point-of-entry: coin enters at its first valid bar (realistic; mirrors when trading would actually begin) ...
symbol metadata, first 10 of 600:

  symbol      first_bar      last_bar  nBars \
0  ETHUSDT 2017-08-17 04:00:00+00:00 2026-06-22 23:00:00+00:00 77437
1  BTCUSDT 2017-08-17 04:00:00+00:00 2026-06-22 23:00:00+00:00 77437
2  BNBUSDT 2017-11-06 03:00:00+00:00 2026-06-22 23:00:00+00:00 75501
3  BCCUSDT 2017-11-11 08:00:00+00:00 2018-11-20 02:00:00+00:00 8805
```

```

4 NEOUSDT 2017-11-20 03:00:00+00:00 2026-06-22 23:00:00+00:00 75165
5 LTCUSDT 2017-12-13 03:00:00+00:00 2026-06-22 23:00:00+00:00 74613
6 QTUMUSDT 2018-03-19 08:00:00+00:00 2026-06-22 23:00:00+00:00 72337
7 ADAUSDT 2018-04-17 04:00:00+00:00 2026-06-22 23:00:00+00:00 71645
8 XRPUSDT 2018-05-04 08:00:00+00:00 2024-06-30 23:00:00+00:00 53905
9 EOSUSDT 2018-05-28 05:00:00+00:00 2025-05-26 02:00:00+00:00 61231

```

```

coverage max_gap_hours halt_flag active
0 0.9984 75 True True
1 0.9984 75 True True
2 0.9984 75 True True
3 0.9815 105 True False
4 0.9984 75 True True
5 0.9984 75 True True
6 0.9988 10 False True
7 0.9988 10 False True
8 0.9984 10 False True
9 0.9986 10 False False

```

Splitting methodology: forward-chained, not stratified

Objective: estimate out-of-sample performance under REGIME DRIFT, not interpolation within a known distribution. The data-generating process drifts across time -- the 2017-18 ICO mania, the 2021 bull run, the 2022 deleveraging, and the current regime are different processes. A split that interleaves test bars among training bars borrows the future to predict the past and yields an error estimate that will not survive live trading.

Stratified sampling is rejected. Stratifying on variance forces similar class proportions across splits by drawing observations regardless of position in time, which destroys the temporal ordering that is the whole object of interest. It answers "can the model interpolate within a known distribution"; the question that matters is "can it extrapolate forward into an unknown regime." Only a forward-chained split measures that.

Design: expanding/rolling walk-forward on shared calendar boundaries; purge/embargo = max(feature lookback, label horizon) at every cut; split on time not on coins; per-fold universe drawn point-in-time from the Stage B profile; transforms fit on train only; per-fold metrics weighted by coin count.

purge/embargo = max(feature lookback 3000, label horizon 48) = 3000 bars = 125 days
walk-forward folds: 5

```

fold train_start train_end test_start test_end n_train n_test \
0 0 2019-04-01 2024-06-23 2024-10-26 2025-02-23 279036 28742
1 1 2019-04-01 2024-10-21 2025-02-23 2025-06-23 302694 25482
2 2 2019-04-01 2025-02-18 2025-06-23 2025-10-21 337209 31086
3 3 2019-04-01 2025-06-18 2025-10-21 2026-02-18 364391 24993
4 4 2019-04-01 2025-10-16 2026-02-18 2026-06-18 395181 16801

```

```

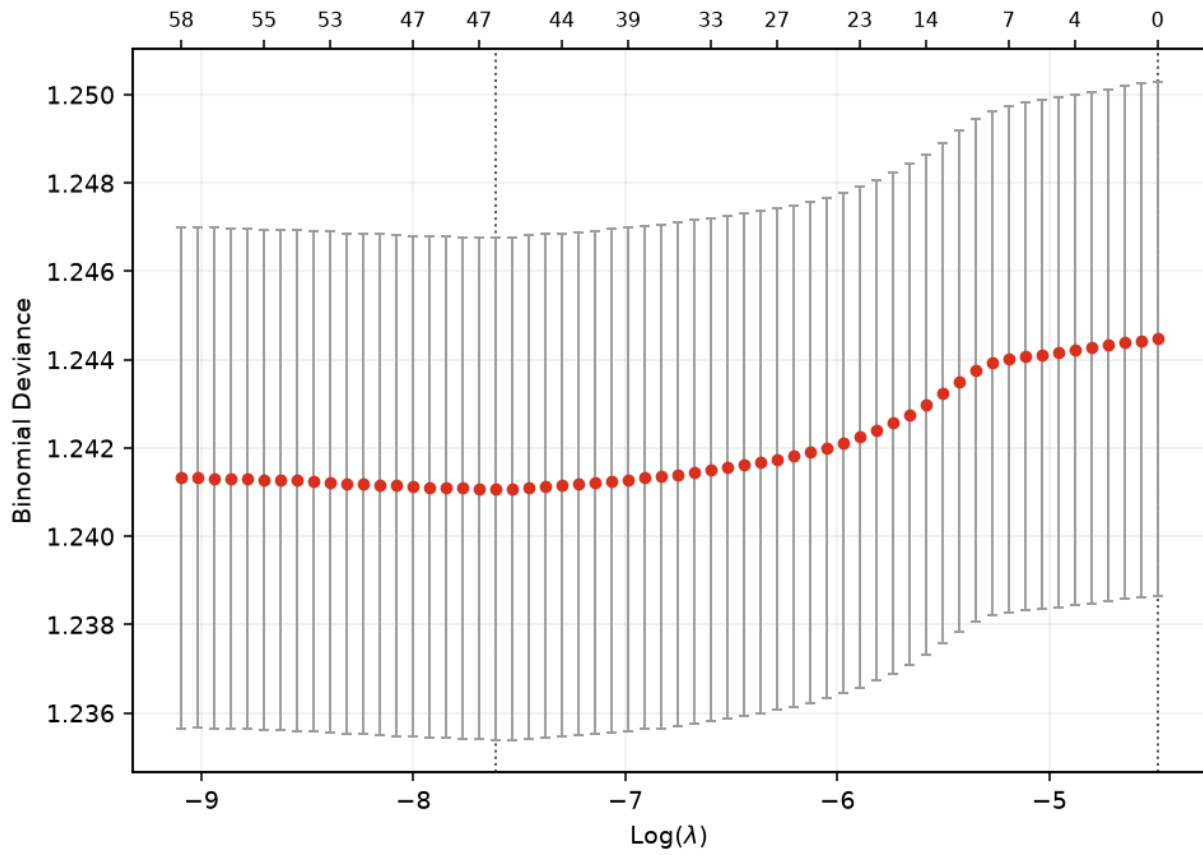
n_coins embargo_days
0 21 125
1 19 125
2 23 125
3 21 125
4 20 125

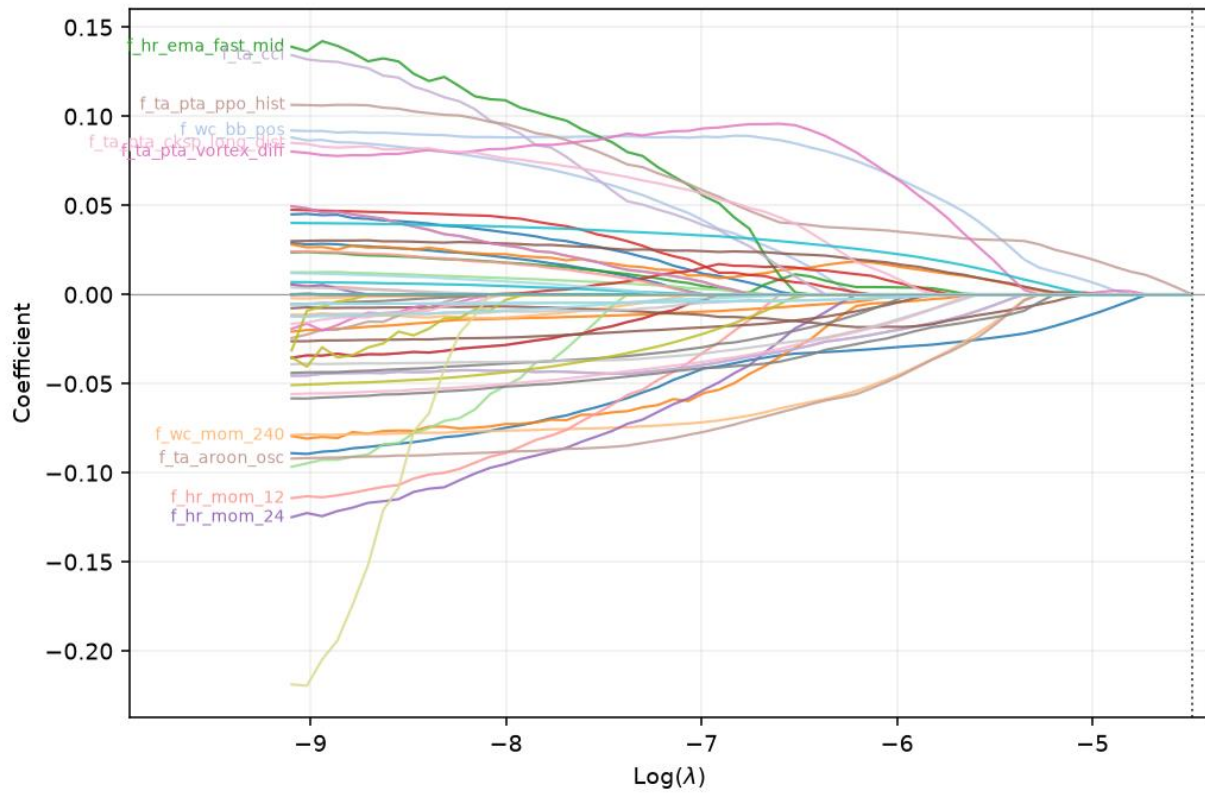
```

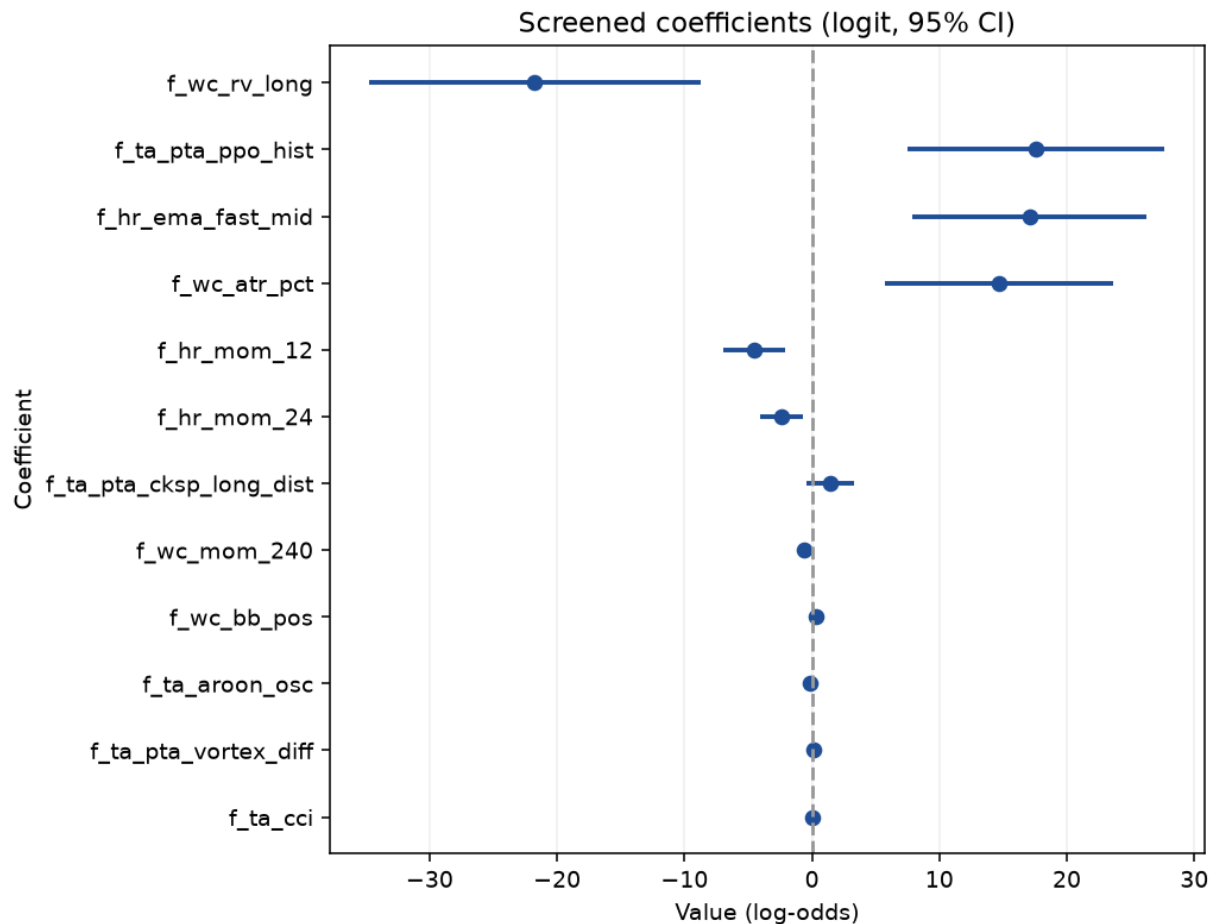
3.6 Variable Selection

Feature selection on the training window only, so the choice never sees the blind year. It ranks the candidate features and prunes the redundant and the near-zero-signal, keeping the causal, scale-invariant set the model trains on, with the after-fee out-of-sample scoreboard as the final arbiter of whether a family earns its place.

lasso path on 25,000 TRAIN rows x 61 features.
lambda.min=0.000494 (nonzero 46), lambda.1se=0.01121 (nonzero 0)







elastic-net kept 46 of 61 features at lambda.1se.

top kept: f_hr_ema_fast_mid(+0.093), f_wc_bb_pos(+0.089), f_ta_aroon_osc(-0.086), f_ta_pta_vortex_diff(+0.086), f_ta_pta_ppo_hist(+0.083), f_hr_mom_24(-0.083), f_wc_mom_240(-0.075), f_hr_mom_12(-0.071), f_ta_cci(+0.071), f_ta_pta_cksp_long_dist(+0.071), f_wc_rv_long(-0.067), f_wc_atr_pct(+0.066), f_wc_ema_fast_mid(-0.065), f_tl_sar_dist(-0.049), f_ta_pta_fisher(-0.046)

interactive coefficient path (open in browser): </Volumes/PortableSSD/Github/day-trader/04-outputs/AA-evals/varselect/coefpath.html>

To train the head-to-head on this subset: set `feat = SELECTED_FEATURES` then re-run training.

Which inputs actually carry signal. Variable selection above asks which columns a penalised regression keeps. This asks a blunter question of the model that is actually used, and answers it three ways, because no single statistic is sufficient alone. A feature's own area under the curve measures how well it ranks winners above losers by itself, where 0.500 is a coin flip. Its stability says in how many of five walk-forward folds it pointed the same way it did over the whole window, so a feature that helps in one stretch of market and hurts in the next is exposed as a regime rather than a feature. Its shuffle cost says how much worse the model scores when that one column is scrambled, which catches a column that is useless alone and valuable beside others, and equally one that duplicates something the model already has. Everything is computed on the training window; the blind final year is never opened.

```
.venv/bin/python 03-inputs/feature_report.py --frames 4h 1d --rows 120000
```

Which inputs carry signal (08 September 2026)

Computed on the training window of each frame. The blind final year was not opened.

Three columns, and none of them is trustworthy alone. **alone** is how far a feature's own AUC sits from 0.500, a coin flip, so 0.020 means it ranks winners above losers 52 per cent of the time by itself. **stable** is the share of walk-forward folds in which it pointed the same way as it did over the whole window, so 0.40 means it changed sides in three folds out of five and is a regime rather than a feature. **in company** is how much worse the model's Brier score gets when that one column is shuffled, which catches a feature that is useless alone and useful beside others, and equally a feature that duplicates something the model already has.

The 4h frame

40,000 training rows to 2025-06-21, 90 features, base rate 0.262.

By family

family	what it is	inputs	alone	stable	in company	missing %
f_btc_	strength relative to the market (bitcoin, or SPY on shares)	7	0.0188	0.83	+0.00692	0.0
f_ta_pta_	pandas-ta indicators	7	0.0486	1.00	+0.00226	0.0
f_w1_	the weekly picture	5	0.0170	0.72	+0.00213	0.0
f_wc_	the coin's own price over day-length windows	12	0.0143	0.67	+0.00172	0.0
f_ta_	in-house indicators (Williams %R, stochastic, CCI, money flow, ADX, Aroon)	8	0.0444	0.90	+0.00160	0.0
f_hr_	the coin's own price over intraday windows	14	0.0256	0.74	+0.00151	0.0
f_d1_	the daily picture	5	0.0077	0.72	+0.00115	0.0
f_flow_	buy and sell order flow	4	0.0085	0.75	+0.00088	0.0
f_st_	triple Supertrend	7	0.0313	0.89	+0.00055	0.0
f_mst_	adaptive Supertrend with the efficiency gate	6	0.0077	0.60	+0.00041	0.0
f_tl_	TA-Lib extras (parabolic SAR, MESA, Hilbert cycle, candle patterns)	15	0.0116	0.75	+0.00029	0.0

The ten the model leans on hardest

feature	family	alone	stable	in company
---------	--------	-------	--------	------------

feature	family	alone	stable	in company
f_btc_mom_168	f_btc_	0.0017	0.80	+0.01810
f_btc_mom_24	f_btc_	0.0367	0.80	+0.01228
f_btc_mom_6	f_btc_	0.0358	1.00	+0.00725
f_ta_aroon_osc	f_ta_	0.0190	0.80	+0.00544
f_ta_pta_cksp_long_dist	f_ta_pta_	0.0616	1.00	+0.00539
f_ta_pta_ppo_hist	f_ta_pta_	0.0567	1.00	+0.00497
f_hr_atr_pct	f_hr_	0.0203	0.60	+0.00484
f_btc_corr_168	f_btc_	0.0112	0.60	+0.00422
f_btc_beta_168	f_btc_	0.0161	1.00	+0.00407
f_ta_pta_trix	f_ta_pta_	0.0308	1.00	+0.00371

The ten the model does not use

Shuffling any of these changed the model's score by less than one part in a hundred thousand, so the model is not reading them at all. 16 of 90 features sit in that band.

feature	family	alone	stable	in company
f_d1_st_up	f_d1_	0.0083	1.00	-0.00000
f_mst_flip	f_mst_	0.0035	1.00	+0.00000
f_tl_cdl_three_black	f_tl_	0.0000	0.33	+0.00000
f_flow_taker_ratio	f_flow_	0.0118	1.00	+0.00000
f_tl_cdl_three_white	f_tl_	0.0001	0.75	+0.00000
f_tl_cdl_eveningstar	f_tl_	0.0004	0.60	+0.00000
f_tl_cdl_morningstar	f_tl_	0.0004	0.60	+0.00000
f_tl_cdl_doji	f_tl_	0.0054	1.00	+0.00000
f_tl_cdl_shootingstar	f_tl_	0.0008	0.80	+0.00000
f_tl_cdl_hammer	f_tl_	0.0045	1.00	+0.00000

14 of 90 features held their direction in fewer than three folds out of five. Those are the ones that describe a period rather than a market.

Full table: 04-outputs/AA-evals/2026-09-08/feature-report-4h-20260908.csv

3.7 Training Regime

The model trains on all history before the final-year cut and is scored once on the blind year. A capped sample keeps the in-notebook fit quick; the full fit runs from the .venv. The section reports

the held-out margin of the model's confident picks over the base rate, the feature importances, and a look at one tree, so the training is legible rather than a black box. The label and horizon are the 0.5 geometry.

```
# Split and scoring come from the scripts: t1.split for the 1h final-year hold-out; tm.evaluate,
# tm.build_models, tm.confidence_filtered shared with 03-inputs/train_model.py. Nothing reimplemented.
train, test, cut = t1.split(df)
Xtr, ytr, Xte, yte = train[feat], train["label"], test[feat], test["label"]
base_te = yte.mean()
span = lambda d: f"{d['datetime'].min().date()} to {d['datetime'].max().date()}"
print(f"train {len(train):,} ({span(train)}) . test {len(test):,} ({span(test)}) . "
      f"cut {pd.Timestamp(cut).date()} embargo +/-{t1.EMBARGO_DAYS}d . base rate {base_te:.3f}")

train 381,207 (2017-12-15 to 2025-06-16) . test 76,789 (2025-06-19 to 2026-06-18) . cut 2025-06-18 embargo +/-2d . base rate 0.303

fitting on 80,000 of 381,207 train rows (2017-12-16 to 2025-06-16) [sampled for the notebook; full run is 03-inputs/train_model_1h.py]

--- LogisticRegression ---
train CV ROC-AUC (5-fold TimeSeriesSplit): 0.528
test accuracy : 0.484
test precision(buy): 0.308 (base rate 0.303)
test recall(buy) : 0.561
test ROC-AUC : 0.504
confusion matrix [rows=true 0/1, cols=pred 0/1]:
[[24148, 29368], [10226, 13047]]
precision lift over base rate: +0.005
confidence filter (act if p>=0.60 or p<=0.40): keeps 2.7% of test rows (n=2099)
precision(buy) 0.281 recall 0.660 F1 0.394 (base rate 0.303)

--- RandomForest ---
train CV ROC-AUC (5-fold TimeSeriesSplit): 0.526
test accuracy : 0.489
test precision(buy): 0.310 (base rate 0.303)
test recall(buy) : 0.557
test ROC-AUC : 0.509
confusion matrix [rows=true 0/1, cols=pred 0/1]:
[[24627, 28889], [10319, 12954]]
precision lift over base rate: +0.007
confidence filter (act if p>=0.60 or p<=0.40): keeps 1.2% of test rows (n=931)
precision(buy) 0.000 recall 0.000 F1 0.000 (base rate 0.303)

--- LightGBM ---
train CV ROC-AUC (5-fold TimeSeriesSplit): 0.518
test accuracy : 0.532
test precision(buy): 0.309 (base rate 0.303)
test recall(buy) : 0.438
test ROC-AUC : 0.509
confusion matrix [rows=true 0/1, cols=pred 0/1]:
[[30706, 22810], [13090, 10183]]
precision lift over base rate: +0.006
confidence filter (act if p>=0.60 or p<=0.40): keeps 27.5% of test rows (n=21084)
precision(buy) 0.294 recall 0.349 F1 0.319 (base rate 0.303)

chosen: RandomForest (precision 0.310 vs base 0.303, AUC 0.509)

# Honesty gate: precision must clearly beat the base rate and AUC clear 0.55. It does not
# trade; Metric 2 (next cell) is the money test.
margin = best["prec"] - base_te
go = (best["prec"] > base_te + 0.05) and (best["auc"] > 0.55)
verdict = "GO" if go else "NO-GO"
print("HONESTY GATE:", "GO (edge survives out-of-sample)" if go
      else "NO-GO (no demonstrable edge - do not trade)")
MODEL_DIR.mkdir(parents=True, exist_ok=True) # self-sufficient: works even if the setup cell was not re-run
joblib.dump({"model": best_model, "features": feat, "name": best_name,
            "trained_through": str(pd.Timestamp(cut).date())},
```

```

"test_base_rate": float(base_te), "go": bool(go)), MODEL_DIR / "model_1h.joblib")
summary = (f"({best_name}): test precision(buy)={best['prec']:.3f} base_rate={base_te:.3f} "
          f"lift={margin:+.3f} AUC={best['auc']:.3f} recall={best['rec']:.3f} acc={best['acc']:.3f} -> {verdict}")
(MODEL_DIR / "model_metrics_1h.txt").write_text(summary + "\n\n" + "\n".join(lines) + "\n")
print("saved", MODEL_DIR / "model_1h.joblib", "and model_metrics_1h.txt")

HONESTY GATE: NO-GO (no demonstrable edge - do not trade)
saved /Volumes/PortableSSD/Github/day-trader/04-outputs/3B-model-training/model_1h.joblib and model_metrics_1h.txt

evaluation record: /Volumes/PortableSSD/Github/day-trader/04-outputs/AA-evals/2026-09-08/eval-head-to-head-20260908.md

# Teaching display: the strongest tree model's top features. The AA-evals record above
# already saves this chart; this is just an inline look.
tree = next((mdl for (_, name, mdl, _) in scored
               if name in ("LightGBM", "RandomForest")), None)
imp = getattr(tree, "feature_importances_", None) if tree is not None else None
if imp is not None:
    order = np.argsort(imp)[::-1][:15]
    fig, ax = plt.subplots(figsize=(8, 5))
    ax.barh([feat[i] for i in order][::-1], imp[order][::-1], color="#0B3D66")
    ax.set_title(f"{tree.__class__.__name__}: top 15 feature importances")
    fig.tight_layout()
    (OUTPUTS / "PNG").mkdir(parents=True, exist_ok=True)
    fig.savefig(OUTPUTS / "PNG" / "3B-feature-importance.png", dpi=160)
    plt.show()

```

3.8 Model Tuning

Tuning asks one question. Did changing a model's settings make it less wrong on data it did not train on, or only on data it did. Two sweeps answer it. The label sweep moves the ATR barrier and the horizon, which changes what the model is asked to predict. The hyperparameter sweep moves the model's own settings over expanding walk-forward folds, which changes how it learns. Both are scored after fees on the out-of-sample scoreboard and both write a dated record, so a later reader can see the before and the after rather than a single number with no history.

The sweeps run full-market as scripts, never inside this document, because a grid of fits takes far longer than a render should.

```

.venv/bin/python 03-inputs/sweep_label_1h.py      # label geometry
.venv/bin/python 03-inputs/walkforward.py         # exit geometry
.venv/bin/python 03-inputs/model_assessment_1h.py --tune # hyperparameters

```

Every sweep on disk is rendered below, newest first, with the incumbent settings marked. The overfit ratio is the cross-validated error divided by the training error. A model that has memorised its training data scores a low training error and a high held-out one, so the ratio climbs; anything above 1.1 is rejected by house rule regardless of how good its held-out error looks.

4h frame, 2026-09-06 . target: bars until the Supertrend flips
panel: crypto 4h, 40 coins, 250,000 rows
9 settings, 3 walk-forward folds each. Scored on the training window only; the blind stretch is untouched.

settings	train	held-out	ratio	verdict
learning_rate=0.03 max_depth=4 max_iter=300 min_samples_leaf=100	18.994	24.159	1.272	REJECTED, overfit
learning_rate=0.03 max_depth=10 max_iter=300 min_samples_leaf=100	15.823	24.362	1.540	REJECTED, overfit
learning_rate=0.03 max_depth=6 max_iter=300 min_samples_leaf=100	16.657	24.399	1.465	REJECTED, overfit
learning_rate=0.06 max_depth=4 max_iter=300 min_samples_leaf=100	17.438	24.564	1.409	REJECTED, overfit
learning_rate=0.06 max_depth=10 max_iter=300 min_samples_leaf=100	13.461	24.752	1.839	REJECTED, overfit
learning_rate=0.06 max_depth=6 max_iter=300 min_samples_leaf=100 (incumbent)	14.382	24.804	1.725	REJECTED, overfit
learning_rate=0.12 max_depth=10 max_iter=300 min_samples_leaf=100	10.772	25.199	2.339	REJECTED, overfit
learning_rate=0.12 max_depth=4 max_iter=300 min_samples_leaf=100	15.413	25.288	1.641	REJECTED, overfit

```
learning_rate=0.12 max_depth=6 max_iter=300 min_samples_leaf=100 11.689 25.526 2.184 REJECTED, overfit
```

Best held-out error 24.159, worst 25.526, so the whole grid spans 1.366.

0 of 9 settings pass the 1.1 overfit bar.

record: 04-outputs/AA-evals/2026-09-06/model-tuning-20260906-1145.json

Reading the sweep above. Two things are worth saying plainly about what tuning has bought so far, because both cut against the instinct to keep turning knobs.

The first is that the settings barely matter. Across the nine settings tried on the four-hour trend-life target, the training error nearly halved while the held-out error got slightly worse. That is the signature of a grid that is buying memorisation, not skill. The winner was the weakest model in the grid, the expected result when the ceiling is set by the data rather than the model.

The second is that the period matters more than the settings. Inside any single setting the three walk-forward folds land between roughly 3.6 and 4.2 bars apart, while the best and worst settings in the whole grid are 1.366 bars apart. Which stretch of market a model is tested on moves the score about three times as much as any knob does. That is the argument for walk-forward evaluation over a single split, and the argument against reading a leaderboard of settings as though the top row had earned its position rather than drawn a favourable fold.

3.9 Model Assessment

The caret-style scorecard, across every built frame. For each of 1h, 4h, and 5m it fits a bounded zoo on a sampled slice and scores a blind final year, reporting Full and cross-validated RMSE on predicted probabilities, the RMSEratio overfit flag, and the after-fee blind read, one block per frame. Frames are scored independently and stacked, never pooled.

RECOMPUTE off -- showing latest persisted full-zoo record: model-assessment-20260906.md

Model assessment (2026-09-06) – caret-style, 1h frame

RMSE/MAE are on predicted probabilities (RMSE = $\sqrt{\text{Brier}}$), the caret-style classification RMSE). Full = in-sample on the training window; CV = time-series out-of-fold on the training window; RMSEratio = CV RMSE / Full RMSE (near 1 = stable, above 1.1 = rejected as overfit). The final year is held out as a single blind test (below). Ranked by CV RMSE, best first.

model	hyperparameters	Full MAE	Full RMSE	CV MAE	CV RMSE	RMSEratio
HistGBM	leaves=31 lr=0.05 iter=600	0.2181	0.2619	0.3861	0.4818	1.840
LightGBM	leaves=31 lr=0.05 n=600	0.2387	0.2806	0.4036	0.4826	1.720
RF	mtry=auto ntree=400 depth=8	0.4445	0.4526	0.4761	0.4840	1.069
LogReg.glm	C=1.0	0.4718	0.4855	0.4463	0.4909	1.011

Best by CV RMSE: HistGBM (CV RMSE 0.4818, RMSEratio 1.840).

Blind final-year test of the best model (scored once): AUC 0.509, precision(buy|p>=0.60) 0.249 vs base rate 0.254, net P&L/trade -0.454% on 3,012 trades.

Classification view (blind test, 0.5 threshold) – diagnostics for whether the minority (barrier-hit) class is learned; the after-fee Metric 2 above remains the GO/NO-GO. Cohen's Kappa 0.006; OOB nan (RandomForest only).

class	precision	recall	F1
0 (no-hit)	0.748	0.706	0.726
1 (barrier-hit)	0.258	0.301	0.278

Confusion matrix [true x pred]: TN 9,942, FP 4,142, FN 3,348, TP 1,440.

3.10 Probability Calibration

Every model in this document emits a probability and is scored on the Brier score, the mean squared distance between that probability and the outcome. Until 8 September 2026 nothing checked whether the probability meant what it said, that is, whether the bars called 30 per cent went on to hit the barrier 30 per cent of the time. A model can rank well and still be badly calibrated, and a miscalibrated probability cannot be used for sizing, because the Kelly fraction takes the probability as an input.

Three statistics are reported. Expected calibration error is the mean absolute gap between predicted and observed frequency, weighted by bin population. Maximum calibration error is the widest gap in any single bin, which matters because a confidence threshold trades inside one bin rather than across the average. Murphy's decomposition splits the Brier score into reliability, the calibration penalty, minus resolution, which rewards departure from the base rate, plus uncertainty, which is the base rate's own variance and belongs to the data rather than the model.

Two post-hoc mappings are fitted on a held-out fifth of the training window and never on the blind year. Platt scaling is a one-parameter logistic fit on the log-odds of the raw probability. Isotonic regression is non-parametric and monotone. Both are monotone, so neither changes AUC or the order in which trades are ranked; they change the number a threshold and a sizing rule read.

```
.venv/bin/python 03-inputs/calibration.py --interval 4h --rows 120000
.venv/bin/python 03-inputs/calibration.py --interval 4h --rows 120000 --unbalanced
```

Probability calibration, 4h frame (08 September 2026)

Fitted on 48,000 training observations to 2025-06-21, calibrated on a held-out 12,000-row slice of the same training window, and scored once on the 18,872-row blind year. Base rate 0.254. The estimator carries `class_weight=balanced`.

A calibrated probability means what it says: of the observations the model called 30 per cent, 30 per cent should have gone on to hit the barrier. Expected calibration error is the average gap between predicted and observed frequency weighted by bin population; maximum calibration error is the widest gap in any bin, which matters because a confidence threshold trades inside one bin rather than across the average.

Summary

mapping	ECE	MCE	Brier	reliability	resolution	uncertainty
---------	-----	-----	-------	-------------	------------	-------------

mapping	ECE	MCE	Brier	reliability	resolution	uncertainty
raw	0.2344	0.7005	0.2586	0.06997	0.00092	0.1893
Platt	0.0295	0.2818	0.1913	0.00257	0.00046	0.1893
isotonic	0.0314	0.6461	0.1925	0.00373	0.00048	0.1893

Brier decomposes as reliability minus resolution plus uncertainty. Reliability is the calibration penalty and falls toward zero as a mapping does its job. Resolution rewards departure from the base rate and is a property of the ranking, so a monotone mapping leaves it close to unchanged. Uncertainty is the base rate's own variance and belongs to the data.

Reliability, raw

predicted band	n	mean predicted	observed frequency	gap
0.0-0.1	14	0.080	0.214	+0.134
0.1-0.2	229	0.169	0.148	-0.020
0.2-0.3	1,299	0.261	0.181	-0.080
0.3-0.4	3,349	0.356	0.227	-0.129
0.4-0.5	5,629	0.452	0.261	-0.191
0.5-0.6	5,092	0.546	0.283	-0.263
0.6-0.7	1,953	0.640	0.267	-0.373
0.7-0.8	614	0.746	0.262	-0.484
0.8-0.9	458	0.847	0.240	-0.607
0.9-1.0	235	0.926	0.226	-0.701

Reliability, Platt

predicted band	n	mean predicted	observed frequency	gap
0.1-0.2	481	0.184	0.154	-0.030
0.2-0.3	12,968	0.262	0.249	-0.013
0.3-0.4	4,712	0.326	0.280	-0.045
0.4-0.5	565	0.442	0.234	-0.209
0.5-0.6	146	0.528	0.247	-0.282

Reliability, isotonic

predicted band	n	mean predicted	observed frequency	gap
0.0-0.1	43	0.019	0.140	+0.121
0.1-0.2	2,163	0.191	0.200	+0.009

predicted band	n	mean predicted	observed frequency	gap
0.2-0.3	12,000	0.279	0.258	-0.022
0.3-0.4	4,428	0.318	0.272	-0.046
0.4-0.5	109	0.465	0.229	-0.236
0.5-0.6	2	0.569	0.500	-0.069
0.6-0.7	6	0.644	0.333	-0.311
0.7-0.8	8	0.746	0.375	-0.371
0.8-0.9	108	0.850	0.204	-0.646
0.9-1.0	5	1.000	0.400	-0.600

Verdict

The raw probabilities carry an expected calibration error of 0.2344 and a maximum of 0.7005. The lowest error after mapping is Platt at 0.0295. Calibration reduces the error by 0.2049 in probability units.

Calibration is monotone, so it does not change AUC or the order in which trades are ranked. It changes the number a confidence threshold and a Kelly fraction read, which is where a miscalibrated probability does its damage.

Full reliability tables: 04-outputs/AA-evals/2026-09-08/calibration-4h-balanced-20260908.csv

Probability calibration, 4h frame (08 September 2026)

Fitted on 48,000 training observations to 2025-06-21, calibrated on a held-out 12,000-row slice of the same training window, and scored once on the 18,872-row blind year. Base rate 0.254. The estimator carries `class_weight=None`.

A calibrated probability means what it says: of the observations the model called 30 per cent, 30 per cent should have gone on to hit the barrier. Expected calibration error is the average gap between predicted and observed frequency weighted by bin population; maximum calibration error is the widest gap in any bin, which matters because a confidence threshold trades inside one bin rather than across the average.

Summary

mapping	ECE	MCE	Brier	reliability	resolution	uncertainty
raw	0.0736	0.6542	0.2060	0.01728	0.00090	0.1893
Platt	0.0301	0.3367	0.1926	0.00381	0.00046	0.1893
isotonic	0.0304	0.6718	0.1919	0.00356	0.00073	0.1893

Brier decomposes as reliability minus resolution plus uncertainty. Reliability is the calibration penalty and falls toward zero as a mapping does its job. Resolution rewards departure from the

base rate and is a property of the ranking, so a monotone mapping leaves it close to unchanged. Uncertainty is the base rate's own variance and belongs to the data.

Reliability, raw

predicted band	n	mean predicted	observed frequency	gap
0.0-0.1	273	0.081	0.117	+0.036
0.1-0.2	4,272	0.162	0.217	+0.055
0.2-0.3	8,020	0.249	0.268	+0.020
0.3-0.4	4,017	0.340	0.277	-0.062
0.4-0.5	1,039	0.442	0.250	-0.192
0.5-0.6	466	0.545	0.255	-0.289
0.6-0.7	286	0.649	0.283	-0.366
0.7-0.8	248	0.748	0.198	-0.550
0.8-0.9	214	0.846	0.192	-0.654
0.9-1.0	37	0.919	0.351	-0.568

Reliability, Platt

predicted band	n	mean predicted	observed frequency	gap
0.1-0.2	501	0.184	0.136	-0.048
0.2-0.3	13,239	0.260	0.255	-0.005
0.3-0.4	4,238	0.328	0.265	-0.063
0.4-0.5	588	0.442	0.257	-0.186
0.5-0.6	270	0.540	0.204	-0.337
0.6-0.7	36	0.623	0.361	-0.262

Reliability, isotonic

predicted band	n	mean predicted	observed frequency	gap
0.0-0.1	58	0.058	0.103	+0.045
0.1-0.2	1,026	0.165	0.168	+0.003
0.2-0.3	11,037	0.268	0.255	-0.013
0.3-0.4	6,298	0.315	0.270	-0.045
0.4-0.5	200	0.406	0.190	-0.216
0.5-0.6	197	0.545	0.203	-0.342
0.6-0.7	20	0.668	0.050	-0.618

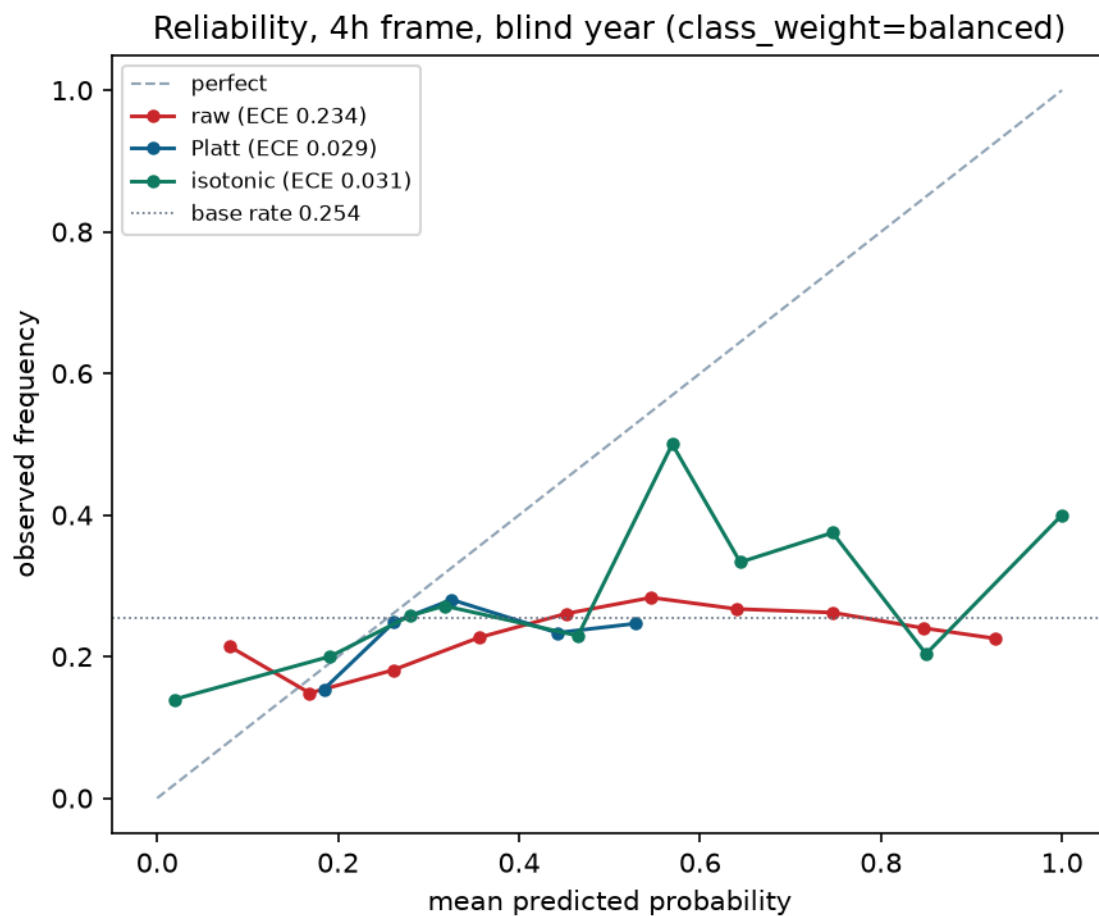
predicted band	n	mean predicted	observed frequency	gap
0.7-0.8	2	0.797	0.500	-0.297
0.8-0.9	11	0.854	0.182	-0.672
0.9-1.0	23	1.000	0.435	-0.565

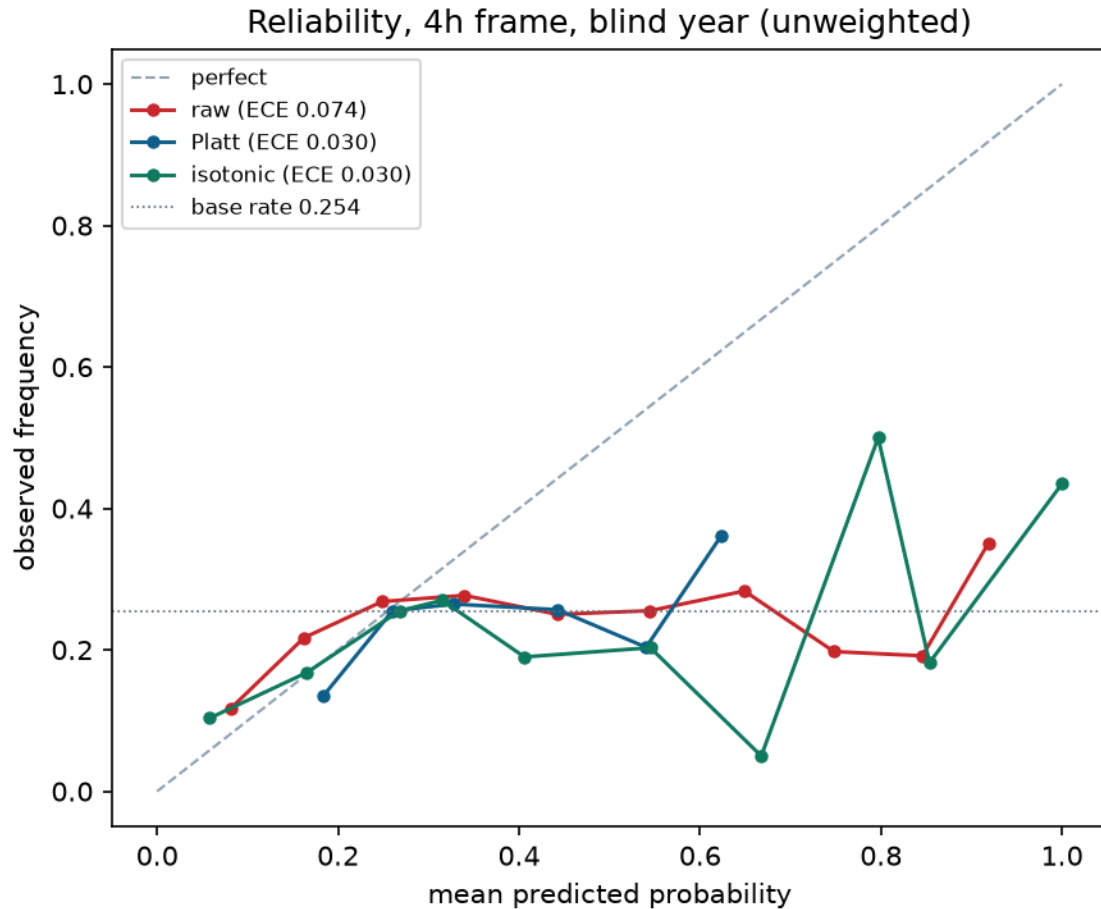
Verdict

The raw probabilities carry an expected calibration error of 0.0736 and a maximum of 0.6542. The lowest error after mapping is Platt at 0.0301. Calibration reduces the error by 0.0435 in probability units.

Calibration is monotone, so it does not change AUC or the order in which trades are ranked. It changes the number a confidence threshold and a Kelly fraction read, which is where a miscalibrated probability does its damage.

Full reliability tables: 04-outputs/AA-evals/2026-09-08/calibration-4h-unweighted-20260908.csv





The result is the largest single improvement in the document. On the four-hour frame the raw probabilities carry an expected calibration error of 0.2344: when the estimator states 92.6 per cent the barrier is hit 22.6 per cent of the time, a gap of 70 percentage points, while the observed frequency moves only between 0.148 and 0.283 across the entire predicted range. Platt scaling reduces the error to 0.0295 and the Brier score from 0.2586 to 0.1913.

The cause is identifiable and it is a choice made in this repository rather than a property of the market. Every estimator in the zoo carries `class_weight="balanced"`, which reweights the minority class and moves predicted probabilities away from the base rate by construction. Refitting the same estimator unweighted cuts the raw calibration error from 0.2344 to 0.0736 and the Brier score from 0.2586 to 0.2060 before any mapping is applied. Class weighting was therefore costing roughly two thirds of the calibration error and a fifth of the Brier score, on the metric these models are ranked by.

Resolution is 0.00092 on the balanced fit, near zero. A calibrated probability is not an edge: the mapping corrects what the numbers mean without adding information the model did not have, and the after-fee out-of-sample scoreboard in 3.14 remains the test.

3.11 Stability

The Monte Carlo robustness gate, per frame. It resamples the model's held-out confident trades ten thousand times to ask whether the result is robust or luck and what the realistic worst case is, reporting total return, drawdown, and Sharpe with their worst-case percentiles, the probability of a loss, and a sign-flip p-value. A frame is ROBUST only if the fifth-percentile total is positive, the loss probability is low, and the p-value small. On the current label every frame reads FRAGILE, which is the expected result for a compounded negative-mean series.

=== [1h] Stability Monte Carlo (reusing loaded frame, 150,000 rows) ===

	actual	P5 (worst)	median	P95
total return	-99.7%	-99.9%	-99.7%	-99.0%
max drawdown	-99.7%	-99.9%	-99.7%	-
Sharpe	-7.38	-9.22	-7.39	-

1h trained model (held-out OOS, confident picks): trades 1511 prob(loss) 100.0% sign-flip p-value 1.0000 -> FRAGILE
monte carlo record: /Volumes/PortableSSD/Github/day-trader/04-outputs/AA-evals/2026-09-08/monte-carlo-20260908-1h.md

=== [4h] Stability Monte Carlo (loaded from disk, 150,000 rows) ===

	actual	P5 (worst)	median	P95
total return	-90.3%	-99.2%	-90.3%	+32.9%
max drawdown	-99.6%	-99.6%	-95.8%	-
Sharpe	-0.65	-2.30	-0.65	-

4h trained model (held-out OOS, confident picks): trades 2010 prob(loss) 92.7% sign-flip p-value 0.7403 -> FRAGILE
monte carlo record: /Volumes/PortableSSD/Github/day-trader/04-outputs/AA-evals/2026-09-08/monte-carlo-20260908-4h.md

=== [5m] Stability Monte Carlo (loaded from disk, 150,000 rows) ===

	actual	P5 (worst)	median	P95
total return	-41.8%	-51.2%	-41.8%	-30.3%
max drawdown	-43.4%	-51.8%	-42.7%	-
Sharpe	-4.84	-6.85	-4.86	-

5m trained model (held-out OOS, confident picks): trades 398 prob(loss) 100.0% sign-flip p-value 1.0000 -> FRAGILE
monte carlo record: /Volumes/PortableSSD/Github/day-trader/04-outputs/AA-evals/2026-09-08/monte-carlo-20260908-5m.md

3.12 Edge Diagnostics

Why the scoreboard returns NO-GO, in three diagnostics plus an out-of-sample selectivity test, run per frame. Q1 asks whether a pre-cost edge exists at all against a base-rate coin flip and a one-bar persistence rule. Q5 asks whether any edge is stable across eras or an artifact of one regime. Q6 sweeps the confidence threshold to test whether selectivity raises expectancy. The out-of-sample selectivity test then chooses the threshold on train and scores it once on the blind year, so a thin

in-sample peak cannot pass. The verdict is a real floor rather than a leakage artefact; the leakage audit is clean.

RECOMPUTE off -- showing the latest persisted records (3 of 10):

Edge diagnostics (2026-09-07) – 1d all-market

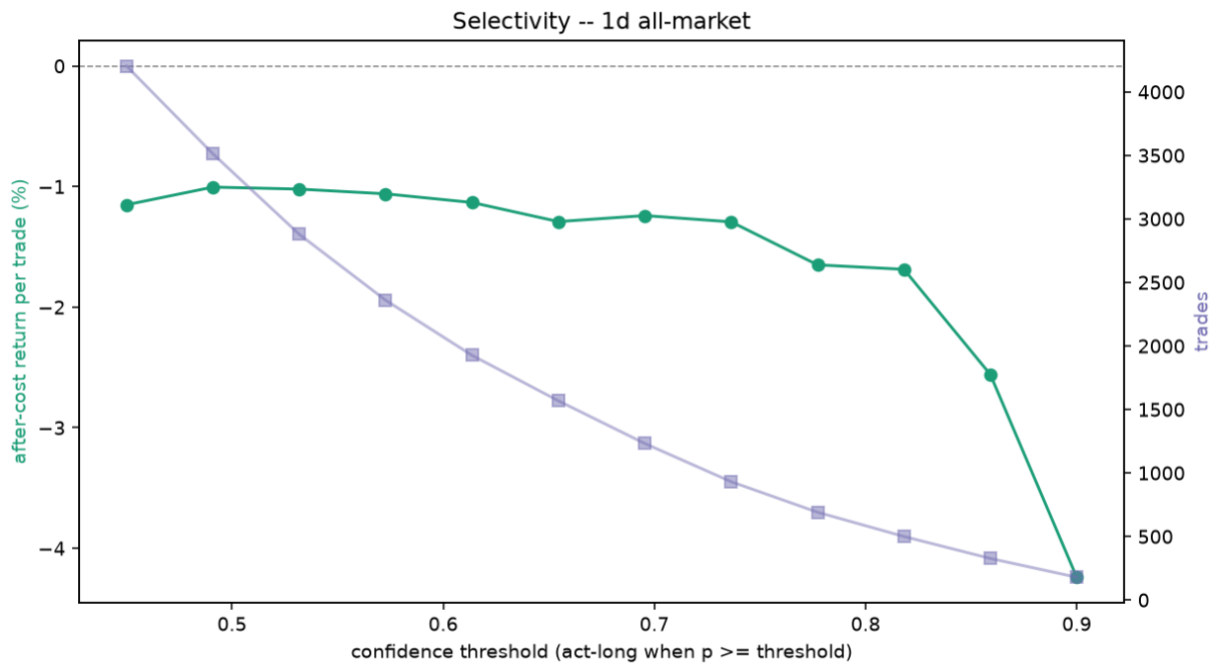
Q1. Pre-cost edge (held-out, fees stripped)

- base rate 0.147 (majority-class accuracy 0.853, informational only)
- model: AUC 0.530 (vs 0.50), accuracy 0.657
- model picks: precision 0.167 vs base 0.147 -> beats the real balance: True
- 1-bar persistence baseline: precision 0.159, pre-cost -2.299%/trade
- model pre-cost mean return per acted trade -1.150% on 418 trades; after-cost -1.350% (cost 0.20% round trip)
- **gate: FAIL - no pre-cost edge; the problem is features/architecture, not costs** (beats base False, beats persistence True)

Q5. Edge stability across eras (after-cost, time-series out-of-fold)

era	trades	after-cost / trade	win rate
2017 launch run-up	0	-	-
2018 bear	0	-	-
2019-20 base	0	-	-
2020-21 bull	0	-	-
2021 top + chop	0	-	-
2022 collapse	1,142	-1.849%	0.22
2023-24 recovery	306	-0.043%	0.34
2025-26 recent	606	-0.473%	0.27

Q6. Selectivity (after-cost return per trade vs confidence threshold)



selectivity

threshold	trades	after-cost / trade	total after-cost	win rate
0.450	4,204	-1.151%	-4838.3%	0.26
0.491	3,513	-1.005%	-3528.9%	0.27
0.532	2,879	-1.021%	-2939.6%	0.26
0.573	2,360	-1.060%	-2502.0%	0.26
0.614	1,926	-1.132%	-2180.1%	0.26
0.655	1,568	-1.291%	-2023.6%	0.25
0.695	1,232	-1.241%	-1528.6%	0.25
0.736	931	-1.293%	-1203.6%	0.25
0.777	691	-1.650%	-1139.9%	0.24
0.818	500	-1.687%	-843.3%	0.24
0.859	327	-2.559%	-836.8%	0.20
0.900	179	-4.240%	-759.0%	0.14

Selectivity reading: after-cost return per trade is flat or falls with the threshold – the confidence ranking carries no usable selectivity; neither a higher threshold nor a retrain on it helps.

Edge diagnostics (2026-09-07) – 4h all-market

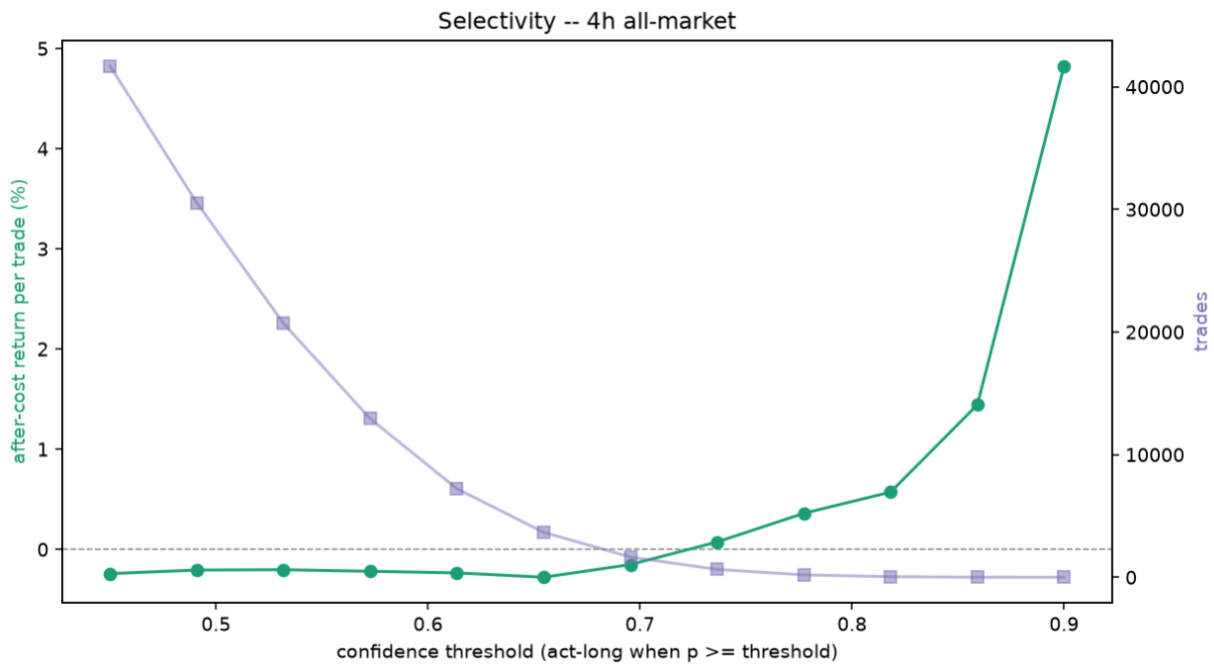
Q1. Pre-cost edge (held-out, fees stripped)

- base rate 0.254 (majority-class accuracy 0.746, informational only)
- model: AUC 0.549 (vs 0.50), accuracy 0.598
- model picks: precision 0.273 vs base 0.254 -> beats the real balance: True
- 1-bar persistence baseline: precision 0.265, pre-cost -0.118%/trade
- model pre-cost mean return per acted trade -0.182% on 1,748 trades; after-cost -0.382% (cost 0.20% round trip)
- **gate: FAIL - no pre-cost edge; the problem is features/architecture, not costs** (beats base False, beats persistence False)

Q5. Edge stability across eras (after-cost, time-series out-of-fold)

era	trades	after-cost / trade	win rate
2017 launch run-up	0	-	-
2018 bear	0	-	-
2019-20 base	0	-	-
2020-21 bull	0	-	-
2021 top + chop	0	-	-
2022 collapse	1,722	-0.262%	0.35
2023-24 recovery	3,493	-0.028%	0.38
2025-26 recent	3,650	-0.316%	0.34

Q6. Selectivity (after-cost return per trade vs confidence threshold)



selectivity

threshold	trades	after-cost / trade	total after-cost	win rate
0.450	41,690	-0.242%	-10096.9%	0.35
0.491	30,545	-0.207%	-6316.9%	0.35
0.532	20,702	-0.203%	-4207.5%	0.35
0.573	12,932	-0.219%	-2832.8%	0.35
0.614	7,217	-0.235%	-1695.0%	0.35
0.655	3,665	-0.278%	-1020.7%	0.35
0.695	1,651	-0.152%	-250.6%	0.36
0.736	640	+0.074%	+47.2%	0.38
0.777	194	+0.360%	+69.9%	0.41
0.818	46	+0.572%	+26.3%	0.41
0.859	4	+1.443%	+5.8%	0.50
0.900	1	+4.825%	+4.8%	1.00

Selectivity reading: after-cost return per trade rises AND crosses zero – at $p \geq 0.777$ it is +0.360% on 194 trades. A selective operating point clears fees: tune that confidence threshold OUT-OF-SAMPLE (walk-forward) with a minimum-trade floor, judged on TOTAL after-cost P&L (best so far at $p \geq 0.777$), not per-trade alone. This is operating-point tuning, not a model retrain.

Edge diagnostics (2026-09-08) – 5m all-market

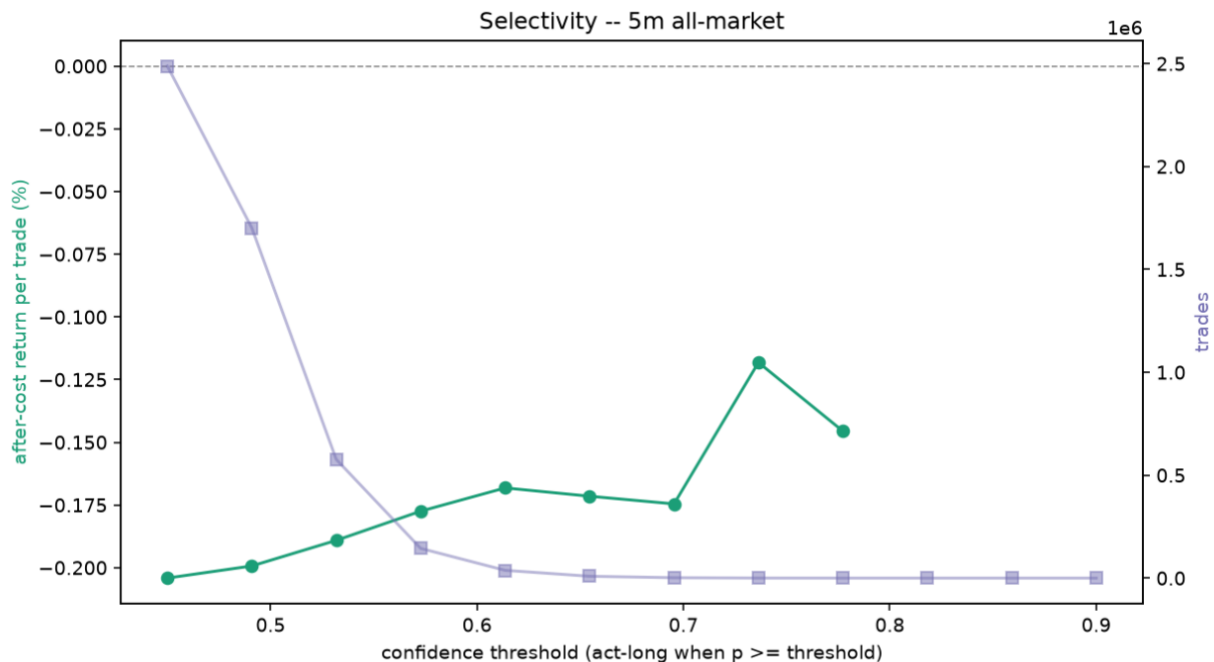
Q1. Pre-cost edge (held-out, fees stripped)

- base rate 0.380 (majority-class accuracy 0.620, informational only)
- model: AUC 0.516 (vs 0.50), accuracy 0.496
- model picks: precision 0.435 vs base 0.380 -> beats the real balance: True
- 1-bar persistence baseline: precision 0.375, pre-cost -0.018%/trade
- model pre-cost mean return per acted trade +0.033% on 1,594 trades; after-cost -0.167% (cost 0.20% round trip)
- **gate: PASS - a pre-cost edge exists to chase** (beats base True, beats persistence True)

Q5. Edge stability across eras (after-cost, time-series out-of-fold)

era	trades	after-cost / trade	win rate
2017 launch run-up	0	-	-
2018 bear	0	-	-
2019-20 base	0	-	-
2020-21 bull	8,390	-0.170%	0.42
2021 top + chop	14,409	-0.165%	0.43
2022 collapse	24,227	-0.174%	0.43
2023-24 recovery	5,924	-0.151%	0.46
2025-26 recent	6,162	-0.192%	0.42

Q6. Selectivity (after-cost return per trade vs confidence threshold)



selectivity

threshold	trades	after-cost / trade	total after-cost	win rate
0.450	2,487,844	-0.204%	-507936.3%	0.39
0.491	1,702,029	-0.199%	-339268.0%	0.40
0.532	576,714	-0.189%	-109068.8%	0.41
0.573	144,223	-0.177%	-25596.2%	0.42
0.614	37,589	-0.168%	-6322.5%	0.43
0.655	8,841	-0.172%	-1516.4%	0.43
0.695	1,760	-0.175%	-307.3%	0.43
0.736	252	-0.118%	-29.8%	0.46
0.777	16	-0.146%	-2.3%	0.44
0.818	0	-	-	-
0.859	0	-	-	-
0.900	0	-	-	-

Selectivity reading: after-cost return per trade rises but stays NEGATIVE at every threshold (-0.204% -> best -0.118% on 252 trades), and the trade count collapses toward the top. The model RANKS well but the frame still loses to fees at every operating point – selectivity does not rescue it. The lever is a coarser decision frame, new information, or cross-sectional framing, NOT a higher threshold and NOT a model retrain on selectivity.

[1h] OOS selectivity on 80,000 of 458,539 rows (reusing loaded frame); threshold on train, scored once...

[4h] OOS selectivity on 80,000 of 3,939,629 rows (loaded from disk); threshold on train, scored once...

[5m] OOS selectivity on 80,000 of 5,529,861 rows (loaded from disk); threshold on train, scored once...

	Frame	threshold (train)	OOS trades	OOS after- cost/trade	OOS total after-cost	OOS win	OOS prec/base	blind ends	verdict
0	1h	0.525	4,229	-0.347%	-1466.2%	0.31	0.304/0.298	2025-06-18	NO-GO (loses to fees out of sample)
1	4h	0.500	6,509	-0.310%	-2016.1%	0.35	0.282/0.254	2025-06-21	NO-GO (loses to fees out of sample)
2	5m	0.550	2,551	-0.198%	-505.4%	0.33	0.414/0.393	2025-06-24	NO-GO (loses to fees out of sample)

3.13 Regime Conditioning

The handoff's Part 2 fix: condition one model on observable regime state rather than switch between per-regime models. The `f_rg` block gives the model trailing volatility and its own-history percentile, trend drift, a Kaufman efficiency, up-down breadth, and the BTC market regime, and the ablation trains the same model with and without it, per frame, comparing cross-era stability and after-fee edge. It carries the same model-performance scoreboard and the Monte Carlo

robustness gate. Conditioning is a generalization fix, more eras profitable and a less bad worst era, not by itself a free edge across the fee line.

RECOMPUTE off -- showing the latest persisted records (3 of 10):

Regime conditioning ablation (2026-09-07) – 4h all-market

Handoff Part 2: one model conditioned on observable regime state (the `f_rg` block) vs the same model without it. Higher after-cost expectancy or a better worst-era (less reliance on one regime) means conditioning helps.

- regime block: 10 features ['f_rg_rv_short', 'f_rg_rv_long', 'f_rg_vol_regime', 'f_rg_drift_short', 'f_rg_drift_long', 'f_rg_efficiency', 'f_rg_updown', 'f_rg_ret_long', 'f_rg_btc_ret_long', 'f_rg_btc_regime']
- features: baseline 90, conditioned 100

model	held-out	after-cost/trade	era mean	era worst	eras positive
baseline (no regime)	AUC 0.549, picks 0.273/base 0.254	-0.382%	-0.202%	-0.316%	0/3
conditioned (+f_rg_)	AUC 0.555, picks 0.282/base 0.254	-0.235%	-0.240%	-0.399%	0/3

Effect of conditioning (two separate questions): cross-era STABILITY does not improve (worst-era -0.083pp, era-mean -0.038pp, eras-positive +0); after-cost EDGE improves (+0.147pp/trade). The point of Part 2 is the first: a model that generalizes across regimes rather than memorizing one. Both remain subject to the after-fee GO/NO-GO bar.

Per-era after-cost (conditioned model)

era	trades	after-cost/trade	win
2017 launch run-up	0	-	-
2018 bear	0	-	-
2019-20 base	0	-	-
2020-21 bull	0	-	-
2021 top + chop	0	-	-
2022 collapse	2,069	-0.399%	0.32
2023-24 recovery	3,503	-0.126%	0.37
2025-26 recent	3,579	-0.195%	0.35

Monte Carlo robustness & model performance (conditioned model, held-out confident trades)

Model performance – the same caret-style scoreboard as the model-assessment table. RMSE and MAE are on predicted probabilities (RMSE = sqrt(Brier)); Full = in-sample on the training window, CV = time-series out-of-fold, RMSEratio = Full / CV (near 1 = stable, well below 1 = overfit). Lower RMSE and higher AUC are better. | metric | baseline | conditioned | | — | — | — | | held-out AUC | 0.5486 | 0.5551 | | held-out RMSE (sqrt Brier) | 0.4844 | 0.4831 | | held-out MAE | 0.4703 | 0.4673 | | Brier score

| 0.2346 | 0.2334 | | Full RMSE (in-sample) | 0.4322 | 0.4280 | | CV RMSE (time-series OOF) | 0.4793 | 0.4793 | | RMSEratio (Full / CV) | 0.902 | 0.893 |

Resampling robustness – 1,807 after-fee per-trade returns x 10,000 sims. ROBUST requires total P5 > 0, p(loss) < 5%, and sign-flip p-value < 0.05. | metric | actual | P5 (worst) | median | P95 | | — | — | — | — | | total return | -99.48% | -99.95% | -99.49% | -94.68% | | max drawdown | -99.72% | -99.96% | -99.63% | - | | Sharpe | -2.97 | -4.64 | -2.98 | - |

- p(loss) **100.0%**, sign-flip p-value **0.9979** -> **FRAGILE**

Regime conditioning ablation (2026-09-08) – 1d all-market

Handoff Part 2: one model conditioned on observable regime state (the `f_rg_` block) vs the same model without it. Higher after-cost expectancy or a better worst-era (less reliance on one regime) means conditioning helps.

- regime block: 10 features ['f_rg_rv_short', 'f_rg_rv_long', 'f_rg_vol_regime', 'f_rg_drift_short', 'f_rg_drift_long', 'f_rg_efficiency', 'f_rg_updown', 'f_rg_ret_long', 'f_rg_btc_ret_long', 'f_rg_btc_regime']
- features: baseline 94, conditioned 104

model	held-out	after-cost/trade	era mean	era worst	eras positive
baseline (no regime)	AUC 0.521, picks 0.173/base 0.147	-1.072%	+1.527%	-3.839%	2/4
conditioned (+f_rg_)	AUC 0.525, picks 0.171/base 0.147	-1.220%	+1.876%	-2.373%	3/4

Effect of conditioning (two separate questions): cross-era STABILITY IMPROVES (worst-era +1.466pp, era-mean +0.349pp, eras-positive +1); after-cost EDGE does not lift (-0.149pp/trade). The point of Part 2 is the first: a model that generalizes across regimes rather than memorizing one. Both remain subject to the after-fee GO/NO-GO bar.

Per-era after-cost (conditioned model)

era	trades	after-cost/trade	win
2017 launch run-up	0	-	-
2018 bear	0	-	-
2019-20 base	0	-	-
2020-21 bull	0	-	-
2021 top + chop	70	+8.923%	0.57
2022 collapse	604	-2.373%	0.21
2023-24 recovery	370	+0.613%	0.35
2025-26 recent	597	+0.342%	0.30

Monte Carlo robustness & model performance (conditioned model, held-out confident trades)

Model performance – the same caret-style scoreboard as the model-assessment table. RMSE and MAE are on predicted probabilities (RMSE = sqrt(Brier)); Full = in-sample on the training window, CV = time-series out-of-fold, RMSEratio = CV RMSE / Full RMSE (near 1 = stable, above 1.1 = rejected as overfit). Lower RMSE and higher AUC are better. | metric | baseline | conditioned | | — | — | — | | held-out AUC | 0.5210 | 0.5252 | | held-out RMSE (sqrt Brier) | 0.4743 | 0.4665 | | held-out MAE | 0.4256 | 0.4212 | | Brier score | 0.2250 | 0.2176 | | Full RMSE (in-sample) | 0.3171 | 0.3084 | | CV RMSE (time-series OOF) | 0.4482 | 0.4475 | | RMSEratio (CV / Full) | 1.413 | 1.451 |

Resampling robustness – 374 after-fee per-trade returns x 10,000 sims. ROBUST requires total P5 > 0, p(loss) < 5%, and sign-flip p-value < 0.05. | metric | actual | P5 (worst) | median | P95 | | — | — | — | | total return | -99.74% | -99.98% | -99.75% | -96.25% | | max drawdown | -99.93% | -99.98% | -99.82% | - | | Sharpe | -2.70 | -4.65 | -2.72 | - |

- p(loss) **100.0%**, sign-flip p-value **0.9963** -> **FRAGILE**

Regime conditioning ablation (2026-09-08) – 4h all-market

Handoff Part 2: one model conditioned on observable regime state (the `f_rg` block) vs the same model without it. Higher after-cost expectancy or a better worst-era (less reliance on one regime) means conditioning helps.

- regime block: 10 features ['f_rg_rv_short', 'f_rg_rv_long', 'f_rg_vol_regime', 'f_rg_drift_short', 'f_rg_drift_long', 'f_rg_efficiency', 'f_rg_updown', 'f_rg_ret_long', 'f_rg_btc_ret_long', 'f_rg_btc_regime']
- features: baseline 90, conditioned 100

model	held-out	after-cost/trade	era mean	era worst	eras positive
baseline (no regime)	AUC 0.549, picks 0.273/base 0.254	-0.382%	-0.112%	-0.378%	1/4
conditioned (+f_rg)	AUC 0.555, picks 0.282/base 0.254	-0.235%	-0.040%	-0.369%	1/4

Effect of conditioning (two separate questions): cross-era STABILITY IMPROVES (worst-era +0.009pp, era-mean +0.072pp, eras-positive +0); after-cost EDGE improves (+0.147pp/trade). The point of Part 2 is the first: a model that generalizes across regimes rather than memorizing one. Both remain subject to the after-fee GO/NO-GO bar.

Per-era after-cost (conditioned model)

era	trades	after-cost/trade	win
2017 launch run-up	0	-	-
2018 bear	0	-	-
2019-20 base	0	-	-
2020-21 bull	0	-	-
2021 top + chop	760	+0.536%	0.45

era	trades	after-cost/trade	win
2022 collapse	2,650	-0.151%	0.36
2023-24 recovery	3,378	-0.175%	0.36
2025-26 recent	3,592	-0.369%	0.33

Monte Carlo robustness & model performance (conditioned model, held-out confident trades)

Model performance – the same caret-style scoreboard as the model-assessment table. RMSE and MAE are on predicted probabilities (RMSE = $\sqrt{\text{Brier}}$); Full = in-sample on the training window, CV = time-series out-of-fold, RMSEratio = CV RMSE / Full RMSE (near 1 = stable, above 1.1 = rejected as overfit). Lower RMSE and higher AUC are better. | metric | baseline | conditioned | | — | — | — | | held-out AUC | 0.5486 | 0.5551 | | held-out RMSE ($\sqrt{\text{Brier}}$) | 0.4844 | 0.4831 | | held-out MAE | 0.4703 | 0.4673 | | Brier score | 0.2346 | 0.2334 | | Full RMSE (in-sample) | 0.4322 | 0.4280 | | CV RMSE (time-series OOF) | 0.4805 | 0.4800 | | RMSEratio (CV / Full) | 1.112 | 1.122 |

Resampling robustness – 1,807 after-fee per-trade returns x 10,000 sims. ROBUST requires total $P5 > 0$, $p(\text{loss}) < 5\%$, and sign-flip p-value < 0.05 . | metric | actual | P5 (worst) | median | P95 | | — | — | — | | total return | -99.48% | -99.95% | -99.49% | -94.68% | | max drawdown | -99.72% | -99.96% | -99.63% | - | | Sharpe | -2.97 | -4.64 | -2.98 | - |

- $p(\text{loss})$ **100.0%**, sign-flip p-value **0.9979** -> **FRAGILE**

3.14 Edge Levers

Added 2026-08-17. Every 4h gate variant lands 5 to 12 basis points short of zero, which locates the problem in the trading frequency, not the filter: each round trip pays the fee, and at 4h cadence the edge cannot cover the toll count. Four levers attack that gap; all are built, and this section renders their records.

1. Coarser frame. The survivorship-complete daily archive (669 pairs, 03-inputs/binance-data/klines/) feeds dataset_1d_allmarket; the baseline build chains straight into the same edge matrix (03-inputs/cross_sectional_regime.py --interval 1d) under both gates and both cost scenarios. Daily bars pay roughly a sixth of the tolls for a similar directional read.
2. Fee engineering. $\text{ACHIEVABLE_COST_PCT} = 0.15$ (maker entry, BNB discount both sides) sits beside the conservative 0.20 default; `trade_binance.fee_status()` measures the account's real rates each run rather than assuming them, and the paper book charges the engineered rate.
3. Narrow book. 03-inputs/edge_attribution.py shows a minority of coins carry all the out-of-sample gated profit (23 of 59 positive, the top five carrying 30pp) while SOL, XRP, and PEPE reliably bleed. 03-inputs/narrow_book.py writes the evidence-based whitelist and the execution layer enforces it.
4. Microstructure features. The opt-in `f_ms_block` (--microstructure, 1d frame only) gives the daily bar hourly eyes: taker-buy flow imbalance and its 7-day mean, up-hour versus down-hour volume, volume concentration, close position in the day's range, realized-versus-range volatility, and 3-sigma hourly jump counts. Causal, scale-invariant, validated on synthetic data; it is built only after the baseline daily edge matrix is read, so the baseline stays clean.

The gates themselves are part of the story: the funding-crowding gate (perpetual funding is mechanically forced positioning information, not derivable from spot price; 03-inputs/funding_features.py)

Gate btc+funding, standard cost 0.20

REGIME-GATED CROSS-SECTIONAL EDGE, 4h frame, gate=btc+funding, after 0.20% cost.
Deciding cell: te_on_top. | signal | gate_open | tr_on_top | tr_on_mkt | te_on_top | te_on_mkt |
te_on_n | te_off_top | te_off_mkt | | — | — | — | — | — | — | — | — | — | — | | f_d1_st_up | 0.47 | -0.313 | -
0.31 | -0.117 | -0.317 | 1820 | -0.165 | -0.438 | | f_wc_rsi | 0.47 | -0.294 | -0.31 | -0.121 | -0.317 | 2965 |
-0.409 | -0.438 | | f_hr_mom_72 | 0.47 | -0.338 | -0.31 | -0.147 | -0.317 | 2965 | -0.291 | -0.438 | |
f_ta_pta_cmo | 0.47 | -0.274 | -0.31 | -0.153 | -0.317 | 2965 | -0.394 | -0.438 | | f_hr_rsi | 0.47 | -0.274 |
-0.31 | -0.153 | -0.317 | 2965 | -0.394 | -0.438 | | f_d1_rsi | 0.47 | -0.300 | -0.31 | -0.160 | -0.317 | 2965 |
-0.437 | -0.438 | | f_wc_mom_60 | 0.47 | -0.307 | -0.31 | -0.182 | -0.317 | 2965 | -0.386 | -0.438 | |
f_d1_mom_10 | 0.47 | -0.319 | -0.31 | -0.202 | -0.317 | 2965 | -0.368 | -0.438 | | f_mst_dir | 0.47 | -
0.441 | -0.31 | -0.203 | -0.317 | 1719 | -0.410 | -0.438 | | f_x_composite | 0.47 | -0.324 | -0.31 | -0.204 |
-0.317 | 3007 | -0.409 | -0.438 | | f_ta_pta_vortex_diff | 0.47 | -0.274 | -0.31 | -0.219 | -0.317 | 2965 | -
0.386 | -0.438 | | f_hr_mom_24 | 0.47 | -0.291 | -0.31 | -0.227 | -0.317 | 2965 | -0.406 | -0.438 |

BEST: f_d1_st_up gate open 47% of bars TEST gate-on : top-third -0.117%/trade (gated market -0.317%) TEST gate-off: top-third -0.165%/trade (market -0.438%) TRAIN gate-on : top-third -0.313%/trade => NOT CLEARED with this gate/cost. Try a stricter gate, the achievable-cost scenario (-cost), or the coarser frame.

REGIME-GATED CROSS-SECTIONAL EDGE, 4h frame, gate=btc+funding, after 0.15% cost.
Deciding cell: te_on_top. | signal | gate_open | tr_on_top | tr_on_mkt | te_on_top | te_on_mkt |
te_on_n | te_off_top | te_off_mkt | | — | — | — | — | — | — | — | — | | f_d1_st_up | 0.47 | -0.263 | -
0.26 | -0.067 | -0.267 | 1820 | -0.115 | -0.388 | | f_wc_rsi | 0.47 | -0.244 | -0.26 | -0.071 | -0.267 | 2965 |
-0.359 | -0.388 | | f_hr_mom_72 | 0.47 | -0.288 | -0.26 | -0.097 | -0.267 | 2965 | -0.241 | -0.388 | |
f_ta_pta_cmo | 0.47 | -0.224 | -0.26 | -0.103 | -0.267 | 2965 | -0.344 | -0.388 | | f_hr_rsi | 0.47 | -0.224 |
-0.26 | -0.103 | -0.267 | 2965 | -0.344 | -0.388 | | f_d1_rsi | 0.47 | -0.250 | -0.26 | -0.110 | -0.267 | 2965 |
-0.387 | -0.388 | | f_wc_mom_60 | 0.47 | -0.257 | -0.26 | -0.132 | -0.267 | 2965 | -0.336 | -0.388 | |
f_d1_mom_10 | 0.47 | -0.269 | -0.26 | -0.152 | -0.267 | 2965 | -0.318 | -0.388 | | f_mst_dir | 0.47 | -
0.391 | -0.26 | -0.153 | -0.267 | 1719 | -0.360 | -0.388 | | f_x_composite | 0.47 | -0.274 | -0.26 | -0.154 |
-0.267 | 3007 | -0.359 | -0.388 | | f_ta_pta_vortex_diff | 0.47 | -0.224 | -0.26 | -0.169 | -0.267 | 2965 | -
0.336 | -0.388 | | f_hr_mom_24 | 0.47 | -0.241 | -0.26 | -0.177 | -0.267 | 2965 | -0.356 | -0.388 |

BEST: f_d1_st_up gate open 47% of bars TEST gate-on : top-third -0.067%/trade (gated market - 0.267%) TEST gate-off: top-third -0.115%/trade (market -0.388%) TRAIN gate-on : top-third -

0.263%/trade => NOT CLEARED with this gate/cost. Try a stricter gate, the achievable-cost scenario (–cost), or the coarser frame.

=== FULL EQUITY BUILD take 2 Tue 18 Aug 2026 05:09:23 UTC built 2547 equities (46 skipped), 4996861 rows, 96 features, base rate 0.253 wrote /Volumes/PortableSSD/Github/day-trader/inputs/alpaca-data/dataset_eq1d_allmarket.parquet === EDGE MATRIX spy+breadth cost 0.05 Tue 18 Aug 2026 05:13:13 UTC equities 1d OOS cut=2025-07-20 00:00:00 gate=spy+breadth (SPY col f_btc_mom_24) cost=0.05% top 10% at 50-name floor candidates=48

EQUITY CROSS-SECTIONAL EDGE, 1d, gate=spy+breadth, after 0.05% cost. Deciding cell:
 te_on_top. | signal | gate_open | tr_on_top | tr_on_mkt | te_on_top | te_on_mkt | te_on_n | te_off_top
 || — | — | — | — | — | — | — | — || f_hr_ema_fast_mid | 0.68 | 0.033 | 0.177 | 0.837 | 0.322 | 19864 |
 0.679 || f_st_ema200_dist | 0.68 | 0.112 | 0.177 | 0.824 | 0.322 | 19864 | 0.668 || f_btc_rel_mom_24 |
 0.68 | 0.045 | 0.177 | 0.812 | 0.322 | 19864 | 0.724 || f_hr_mom_24 | 0.68 | 0.045 | 0.177 | 0.812 |
 0.322 | 19864 | 0.724 || f_wc_mom_20 | 0.68 | 0.080 | 0.177 | 0.798 | 0.322 | 19864 | 0.709 ||
 f_wc_ema_fast_mid | 0.68 | 0.073 | 0.177 | 0.780 | 0.322 | 19864 | 0.417 || f_w1_ema_fast_slow |
 0.68 | 0.181 | 0.177 | 0.753 | 0.322 | 19864 | 0.614 || f_ta_pta_trix | 0.68 | 0.073 | 0.177 | 0.752 | 0.322
 | 19864 | 0.407 || f_mst_dist | 0.68 | 0.067 | 0.177 | 0.744 | 0.322 | 19864 | 1.092 || f_hr_mom_72 |
 0.68 | 0.177 | 0.177 | 0.738 | 0.322 | 19864 | 0.554 || f_btc_rel_mom_168 | 0.68 | 0.169 | 0.177 | 0.737
 | 0.322 | 19864 | 0.591 || f_hr_mom_168 | 0.68 | 0.169 | 0.177 | 0.737 | 0.322 | 19864 | 0.591 |

20 rows in the record, 12 shown.

BEST: f_hr_ema_fast_mid gate open 68% TEST gate-on : top-decile +0.837%/trade (gated market +0.322%) TRAIN gate-on : top-decile +0.033%/trade => CANDIDATE: clears after cost, train/test consistent. Next: the walk-forward kill harness (60% fold pass, tradeable width, attribution breadth). === EDGE MATRIX spy+breadth cost 0.10 (stress) equities 1d OOS cut=2025-07-20 00:00:00 gate=spy+breadth (SPY col f_btc_mom_24) cost=0.1% top 10% at 50-name floor candidates=48

EQUITY CROSS-SECTIONAL EDGE, 1d, gate=spy+breadth, after 0.10% cost. Deciding cell:
 te_on_top. | signal | gate_open | tr_on_top | tr_on_mkt | te_on_top | te_on_mkt | te_on_n | te_off_top
 || — | — | — | — | — | — | — | — || f_hr_ema_fast_mid | 0.68 | -0.017 | 0.127 | 0.787 | 0.272 | 19864 |
 0.629 || f_st_ema200_dist | 0.68 | 0.062 | 0.127 | 0.774 | 0.272 | 19864 | 0.618 || f_btc_rel_mom_24 |
 0.68 | -0.005 | 0.127 | 0.762 | 0.272 | 19864 | 0.674 || f_hr_mom_24 | 0.68 | -0.005 | 0.127 | 0.762 |
 0.272 | 19864 | 0.674 || f_wc_mom_20 | 0.68 | 0.030 | 0.127 | 0.748 | 0.272 | 19864 | 0.659 ||
 f_wc_ema_fast_mid | 0.68 | 0.023 | 0.127 | 0.730 | 0.272 | 19864 | 0.367 || f_w1_ema_fast_slow |
 0.68 | 0.131 | 0.127 | 0.703 | 0.272 | 19864 | 0.564 || f_ta_pta_trix | 0.68 | 0.023 | 0.127 | 0.702 | 0.272
 | 19864 | 0.357 || f_mst_dist | 0.68 | 0.017 | 0.127 | 0.694 | 0.272 | 19864 | 1.042 || f_hr_mom_72 |
 0.68 | 0.127 | 0.127 | 0.688 | 0.272 | 19864 | 0.504 || f_btc_rel_mom_168 | 0.68 | 0.119 | 0.127 | 0.687
 | 0.272 | 19864 | 0.541 || f_hr_mom_168 | 0.68 | 0.119 | 0.127 | 0.687 | 0.272 | 19864 | 0.541 |

20 rows in the record, 12 shown.

BEST: f_hr_ema_fast_mid gate open 68% TEST gate-on : top-decile +0.787%/trade (gated market +0.272%) TRAIN gate-on : top-decile -0.017%/trade => FRAGILE: clears on TEST only; treat as unproven. === EDGE MATRIX ungated cost 0.05 (reference) equities 1d OOS cut=2025-07-20 00:00:00 gate=none (SPY col f_btc_mom_24) cost=0.05% top 10% at 50-name floor candidates=48

```

EQUITY CROSS-SECTIONAL EDGE, 1d, gate=none, after 0.05% cost. Deciding cell: te_on_top. |
signal | gate_open | tr_on_top | tr_on_mkt | te_on_top | te_on_mkt | te_on_n | te_off_top | | — | — | —
| — | — | — | — | — | | f_mst_dist | 1.0 | 0.272 | 0.372 | 0.854 | 0.461 | 29040 | NaN | |
f_hr_ema_fast_mid | 1.0 | 0.245 | 0.372 | 0.787 | 0.461 | 29040 | NaN | | f_btc_rel_mom_24 | 1.0 |
0.241 | 0.372 | 0.784 | 0.461 | 29040 | NaN | | f_hr_mom_24 | 1.0 | 0.241 | 0.372 | 0.784 | 0.461 | 29040
| NaN | | f_st_ema200_dist | 1.0 | 0.247 | 0.372 | 0.775 | 0.461 | 29040 | NaN | | f_wc_mom_20 | 1.0 |
0.264 | 0.372 | 0.770 | 0.461 | 29040 | NaN | | f_st_dist_1 | 1.0 | 0.334 | 0.372 | 0.746 | 0.461 | 29040 |
NaN | | f_mst_mult | 1.0 | 0.321 | 0.372 | 0.723 | 0.461 | 29055 | NaN | | f_mst_ker_rank | 1.0 | 0.327 |
0.372 | 0.720 | 0.461 | 29050 | NaN | | f_w1_ema_fast_slow | 1.0 | 0.253 | 0.372 | 0.709 | 0.461 | 29040
| NaN | | f_btc_beta_168 | 1.0 | 0.521 | 0.372 | 0.707 | 0.461 | 29040 | NaN | | f_hr_mom_168 | 1.0 |
0.343 | 0.372 | 0.691 | 0.461 | 29040 | NaN |

```

20 rows in the record, 12 shown.

BEST: f_mst_dist gate open 100% TEST gate-on : top-decile +0.854%/trade (gated market +0.461%)
 TRAIN gate-on : top-decile +0.272%/trade => CANDIDATE: clears after cost, train/test consistent.
 Next: the walk-forward kill harness (60% fold pass, tradeable width, attribution breadth). ===
 DONE Tue 18 Aug 2026 05:17:40 UTC

narrow book not yet generated: .venv/bin/python 03-inputs/narrow_book.py

4. Shared Method

The machinery shared by every part, the discipline that keeps the three notebooks telling one story and keeps the picture from drifting from the numbers.

4.1 Multi-Frame Loop

The same search runs on three decision frames, 1h, 4h, and 5m, and each is a whole system with its own bars, its own label horizon, its own features, and its own screen, all retuned in place by one builder. The frames are compared, never pooled: a five-minute label and a four-hour label do not mean the same thing, and stacking their rows would blend incompatible targets. Every modelling section in Part 3 from 3.9 on loops over the three frames, scores each on its own out-of-sample, restores the active frame, and reports one block per frame. Cross-timeframe information still reaches each frame through causal higher-timeframe features, so a one-hour model sees the four-hour and daily trend without the datasets being merged.

4.2 After-Fee Scoreboard

One test decides everything. A candidate is scored on a blind final year held out from all training and tuning, and it counts only if it beats a coin flip at the real base rate, beats buy and hold, and returns more than zero per trade after the round-trip fee of about 0.20 percent. The fee is the adversary: a rule that ranks trades well but loses to the fee is a NO-GO. Every sweep, every model, and every ablation reports to this same scoreboard, logged under 04-outputs/AA-evals/, so results across the notebooks are commensurable and no in-sample number can override the out-of-sample verdict.

4.3 Embargoed Splits

Financial bars are autocorrelated, so a random split leaks the future into the past. The evaluation cuts time-ordered: train on all history before a final-year cut, embargo a band equal to the label horizon across the cut so no label straddles it, and score the held-out year once. Cross-validation inside the training window walks expanding time-series folds, never random k-fold. Stage C generalizes this to a forward-chained walk-forward with per-fold point-in-time universes, so a fold only ever sees coins that would have passed the screen as of that bar.

4.4 Reproducibility

Single sources of truth keep the story consistent. The exit mechanics are one module, `exit_geometry_viz`, shared verbatim by all three notebooks, so the candles and the scoreboard cannot disagree. The triple-Supertrend is one benchmark across the chapters and one feature family in the model. Datasets and the flow table are Parquet, columnar and dtype-preserving, because round-tripping timestamps through CSV silently broke the split. The build is offline and deterministic from the checksummed archives, and every evaluation writes a dated record, so a result can be traced to the run that produced it.

5. Findings

The honest state of the search as of this revision. Nothing has cleared the after-fee bar yet; what follows is where each frame lands and where the edge, such as it is, actually lives.

5.1 Current State

Every crypto frame is rejected, and the diagnosis changed on 7 September 2026. Transaction cost had been treated as the binding constraint. The edge diagnostics run that day computed expectancy gross of cost, and both surviving frames were negative. On four-hour bars the model's acted trades returned -0.182 per cent gross, against -0.118 per cent for a one-bar persistence baseline, so the model underperformed the naive benchmark. On daily bars gross expectancy was -1.150 per cent. Both records record the same verdict: no pre-cost edge exists, so the constraint is the feature set and the architecture rather than the cost.

The remaining results are unchanged. On the one-hour frame the label sweeps, the seven-model comparison and the Monte Carlo robustness gate are all rejected at an AUC near 0.50, and the default barrier carries negative unconditional expectancy, a base rate of 0.313 against a breakeven of 0.333. The four-hour frame reaches AUC 0.549, and higher-timeframe context raises it further without clearing the cost. The five-minute frame carries positive gross expectancy but the realised move is smaller than the 0.20 per cent round trip. None of this is attributable to leakage: the label alignment, scaling, feature construction and split code were audited line by line and are clean.

5.2 What Tuning Showed

The hyperparameter sweep was run once, on 6 September 2026. It does not move the result.

Nine hyperparameter combinations were scored on the four-hour trend-life target over three walk-forward folds. All nine failed the overfit criterion. The minimum RMSE ratio in the grid was 1.272 against a rejection threshold of 1.1, and the maximum was 2.339. Across the grid the training RMSE fell from 18.994 to 10.772 bars while the cross-validated RMSE rose from 24.159 to 25.526, the

signature of increasing variance without a gain in generalisation. The combination with the lowest cross-validated error also had the highest training error, a learning rate of 0.03 at depth 4.

The second result is the more consequential. Within any single configuration the three folds span 3.6 to 4.2 bars. Across the entire grid, best to worst configuration, the span is 1.366 bars. Fold-to-fold variance therefore exceeds configuration-to-configuration variance by roughly a factor of three, so a configuration leaderboard read without its fold dispersion ranks sampling noise.

Two defects in model selection were identified and corrected on 8 September. Selection minimised cross-validated error without reference to the overfit ratio, which returned a model at 1.840 against a rejection threshold of 1.1 and then scored the held-out year with it. Separately, the tuner computed no training error, so no overfit ratio existed for a tuning grid and the criterion could not be applied. Both now apply the threshold before reading the ranking, and both report explicitly when no configuration passes.

A separate comparison bounds the return to a richer architecture. Ninety engineered features scored 24.213 bars RMSE on the same target; the raw twenty-four most recent returns, with no feature construction, scored 24.247. The two representations differ by 0.034 bars, so they carry equivalent information and a sequence model has no residual gap to exploit on this target.

5.3 Which Features

The record is 04-outputs/AA-evals/2026-09-08/feature-report-20260908.md, computed on 40,000 four-hour training observations to 21 June 2025, ninety features, base rate 0.262. All estimates are from the training split; the held-out year was not opened. Three statistics are reported. Univariate AUC measures the feature's own discrimination. Sign stability is the proportion of five walk-forward folds in which the univariate AUC falls on the same side of 0.5 as it does over the full window. Permutation importance is the increase in Brier score when that column alone is randomly permuted, which identifies features that are uninformative marginally but contribute conditionally.

Relative strength dominates own-price features. The highest-ranked family is relative strength against bitcoin, `f_btc_`, at a mean permutation importance of 0.0069, roughly three times the next family. Four of the ten highest-importance columns belong to it, led by `f_btc_mom_168`, the 168-hour relative move, at 0.0181, more than twice any other single feature. It is the only family not derived from the asset's own price series. This corroborates the cross-sectional result from an independent direction, as a property of the fitted model rather than of a backtest.

The classical indicator blocks are the most stable. The `f_ta_pta_` block preserved sign in five folds of five for every feature and carries the largest mean univariate AUC deviation of any family at 0.0486. The in-house block follows at 0.0444 with 0.90 sign stability. Neither is large enough to trade in isolation, but both are stable across folds.

The candlestick patterns carry no information. Eight of the ten zero-importance columns are candlestick patterns from the TA-Lib block. `f_tl_cdl_three_black` has a univariate AUC of exactly 0.500 and preserves sign in one fold of three. Permuting any of the sixteen zero-importance columns changes the Brier score by less than $1e-5$. The fifteen-feature TA-Lib family ranks second from bottom on mean permutation importance.

The adaptive Supertrend does not survive multivariate conditioning. `f_mst_` was recorded in June 2026 as the most robust cross-sectional signal in the search. On these statistics it has the lowest

sign stability of any family at 0.60 and ranks second from bottom on permutation importance at 0.0004. Its flip indicator, `f_mst_flip`, is among the zero-importance columns. This does not overturn the cross-sectional result, which was estimated under a different design, but the signal does not retain weight when the model has ninety alternative columns available.

Fourteen of ninety features preserved sign in fewer than three folds of five. These are regime-specific rather than persistent, and they supply the variance that a hyperparameter sweep converts into memorisation. Taken with 5.2 this accounts for the sweep result: with roughly a sixth of the feature set reversing sign between folds, additional model capacity fits noise.

5.4 The Lead

One signal is genuinely present, and 5.3 is now the second independent measurement of it. Ranking the universe by relative strength and holding the top third shows a real and broad edge, with most momentum and trend signals producing a positive top-minus-bottom spread that holds from training into test, and the top third beating the market for a large majority of them. It is not a long-only GO. The universe's own after-fee baseline is about minus 0.38 per cent per trade, so the best top third beats the market while staying below zero. The edge exists and is stable; the negative baseline sinks it.

Two things were never tried on it. Holding longer, so the fee is paid fewer times, and running the same ranking on shares, where the market drifts up rather than down. The second of those is section 5.5.

5.5 The Equity Result

The only rule this project has ever cleared is on shares, not coins, and it is the plainest one in the literature. Rank every name by its return over the previous twelve months, ignore the most recent month, buy the top tenth, hold for a month, repeat without overlapping. Nothing in it is chosen using this project's own model scores, which is what makes it a test rather than a fit.

Over 116 months to 5 September 2026, after a 5 basis point monthly cost, the top tenth returned 2.114 per cent a month against the market's 1.107, a gap of 1.006 percentage points. The gap is 2.41 times its own standard error, and above about two is the conventional bar for calling a result real. It paid in 85 per cent of half-year blocks and beat the market in 80, both above the 60 per cent the harness demands. The same harness killed the six-month version at 55 per cent and killed low volatility at 25 per cent, exactly as the published literature says it should, which is the evidence that the harness can return a negative.

Two caveats belong here rather than in a footnote. The share history covers only companies still trading today, so every bankruptcy is missing and the figure is an upper bound; a separate stress test that injects delistings still cleared the bar. And the most recent half-year block is the worst in the entire series, minus 5.646 points a month, following the best at plus 5.319. The live paper book was funded into the second of those.

5.6 Next Steps

The order follows from 5.2 and 5.3 rather than from the earlier lever list.

First, stop tuning the crypto model. Nine settings moved the held-out error by 1.366 bars while the fold changed it by nearly four, and both surviving frames lose before fees. There is nothing in the settings to find.

Second, cut the inputs that carry nothing. Sixteen features the model does not read, fifteen TA-Lib columns worth 0.0003 between them, and fourteen features that change sign between folds. A smaller, steadier input set is the one change 5.2 and 5.3 both point at.

Third, calibrate. Every model in the zoo trains with `class_weight="balanced"`, which pushes predicted probabilities away from the true base rate, while the headline score is the Brier score, which measures precisely that distance. There is no reliability curve anywhere in the repository and no Platt or isotonic step. Until that exists, no probability this project prints has been checked against how often the thing actually happened.

Fourth, run the ranking on shares. It is the untried half of 5.4 and it sits on the one market where the baseline is positive.

The live switch stays off throughout. None of this changes real exposure.

Appendices

A. File Map

Which module powers which section.

Module	Powers
03-inputs/build_dataset_1h.py	the frame builder: features, label, screen, configure per frame (3.1, 3.2)
03-inputs/acquire_vision.py	Stage A, survivorship-complete acquisition from <code>data.binance.vision</code> (3.5)
03-inputs/profile_panel.py	Stage B, coverage, gap, and liquidity profiling and the point-in-time universe (3.5, and the Part 1 and 2 universe)
03-inputs/wf_splitter.py	Stage C, the forward-chained walk-forward splitter (3.5)
03-inputs/train_model_1h.py	the train and test split, the loader, and the shared evaluation (3.7)
03-inputs/train_model.py	the shared cost and confidence constants and the eval helper
03-inputs/variable_selection.py	feature selection on the training window (3.6)
03-inputs/feature_report.py	which inputs carry signal, alone, across folds and in company (3.6, 5.3)
03-inputs/model_assessment_1h.py	the caret-style scorecard, the model zoo, and the tuner (3.8, 3.9)
03-inputs/calibration.py	reliability, calibration error, Brier decomposition, Platt and isotonic mapping (3.10)
03-inputs/sweep_label_1h.py	the label-geometry sweep (3.8)

Module	Powers
03-inputs/monte_carlo_1h.py	the Monte Carlo robustness gate (3.11)
03-inputs/edge_diagnostics.py	the Q1, Q5, and Q6 diagnostics and the out-of-sample selectivity test (3.12)
03-inputs/regime_conditioning.py	the regime-conditioning ablation (3.13)
03-inputs/cross_sectional_4h.py	cross-sectional tercile ranking (1.2, 5.2)
03-inputs/exit_geometry_1h.py	the exit-geometry sweep (2.4, 3.3)
03-inputs/exit_geometry_viz.py	the shared exit and entry-context visualization (1.6, 2.x, 3.3)
03-inputs/baseline_supertrend_1h.py	the triple-Supertrend after-fee baseline (1.x, 3.11)
03-inputs/walkforward.py	the MACD signal-line cross-up entry experiment (1.3)
03-inputs/entry_sharpening_4h.py	the entry-sharpening tests (3.4)
03-inputs/binance-data/flow_data.py	the trade-flow imbalance data
03-inputs/config.py	operator configuration and the LIVE_TRADING switch
03-inputs/cross_sectional_regime.py	the regime-gated cross-sectional edge matrix (3.14, 5.1)
03-inputs/edge_attribution.py	per-coin attribution of the gated edge (3.14)
03-inputs/narrow_book.py	the evidence-based tradeable whitelist (3.14)
03-inputs/candidate_screen.py	the live fee-adjusted range screen (3.14)
03-inputs/fetch_funding.py, 03-inputs/funding_features.py	the funding archive and the f_fund_ crowding gate (3.14)
03-inputs/portfolio_backtest.py	the gated composite portfolio backtest (3.14)
03-inputs/trade_binance.py	guarded execution: maker entries, measured fees, the narrow-book filter
03-inputs/paper_trade.py	the paper book: hard rules enforced, engineered fee rate (D.2)

B. Glossary

Term	Meaning
Label	the supervised target: did a coin's trade hit the profit barrier before the stop within the horizon
Triple barrier	a label set by an upper (take-profit), a lower (stop), and a time barrier, here ATR-scaled
ATR	average true range, a coin's typical bar-to-bar move, used to scale stops and the volatility band
Supertrend	an ATR-channel trailing stop; three of them behind an EMA-200 gate form the aligned baseline
Kelly	growth-optimal position size given an edge; the book

Term	Meaning
	sizes at half-Kelly to buffer estimation error
Base rate	the unconditional share of positive labels, the coin-flip a model must beat
Breakeven	the win rate at which a fixed-payoff trade nets zero after fees
Embargo	a gap around the train and test cut so no label straddles it and leaks
RMSEratio	Full RMSE over cross-validated RMSE; near 1 generalizes, well below 1 overfits
PPO	the percentage price oscillator, a scale-free MACD used as a feature
Cross-sectional	ranking coins against each other each bar, versus predicting each in isolation
Regime state	observable market context (volatility, trend efficiency, BTC regime) the model conditions on
Corwin-Schultz	a high-low estimator of the bid-ask spread, used where the archives carry no order book
Survivorship-complete	a universe that includes every coin ever listed, delisted ones included
Point-in-time	a screen replayed as of each bar, so a row is kept only if it would have passed then
Walk-forward	forward-chained train-then-test folds that respect time order
GO / NO-GO	the after-fee out-of-sample verdict that gates everything

C. Environment

The repository uses MacPorts, never Homebrew; install with `sudo port install` or a language-native installer, and the pipeline runs from the project `.venv`, a MacPorts Python, with `pandas-ta` and `TA-Lib` pip-installed there. Note the two `pandas-ta` forks: the scan chapters import `pandas-ta-classic` while the model builder imports `pandas-ta`, so the consolidated `venv` needs both. The notebook needs Python 3.11 or newer, and the environment cells install only the libraries each part imports that are missing, to avoid a `numpy` and `numba` resolver conflict. `NumPy` must stay below 2.3 (pinned 2026-08-17): `numba`, which `pandas-ta` imports, refuses `NumPy` 2.4, and the failure is silent feature loss in the dataset build, not a crash.

The repository sits on an exFAT SSD, which cannot store extended attributes, so macOS scatters `._`-prefixed `AppleDouble` files on every write. They are git-ignored but can crash `matplotlib` with a `0xb0` decode error; clear them with a `find ... -name '._*' -delete` if an import throws it. The durable fix is to keep the repository on an APFS or HFS+ volume.

Secrets live in the run environment, never in the repository and never in a `.env` file.

Variable	Purpose
BINANCE_API_KEY, BINANCE_API_SECRET	Binance access
BINANCE_TESTNET	true uses the testnet; false is real funds
ALPACA_API_KEY, ALPACA_API_SECRET, ALPACA_BASE_URL	Alpaca, the later-stage equities venue
LIVE_TRADING	the money switch; false until a strategy clears the bar, and never armed by the model

D. Committee Layer

Added 2026-08-16. A qualitative research layer now sits beside the quant pipeline: the TradingAgents multi-agent committee (TauricResearch, v0.3.1, built and validated in its own venv at /Volumes/PortableSSD/Github/TradingAgents) and a local paper-trading engine that rehearses execution against live prices. Neither places real orders.

D.1 Committee Research

03-inputs/ta_research.py runs each symbol through the committee in crypto mode: market, sentiment and news analysts, a bull and bear debate, a three-way risk panel, and a portfolio manager who returns one of five ratings (Buy, Overweight, Hold, Underweight, Sell) with a written thesis. Models: Claude Sonnet 5 in the debate seats, Haiku 4.5 for quick tasks; alpha in the reflection loop is benchmarked to BTC-USD, not SPY. Ratings and executive summaries append to 05-research/memory/research-log.md; the framework's own decision log, which auto-grades each call against the realised five-day return on the next same-ticker run, lives in 05-research/memory/ta-decisions.md; full report trees land under 04-outputs/ta-reports/.

The committee automates Principle 1 (every entry needs an affirmative case and a devil's advocate) and adds the information channels the price-only pipeline cannot see: news, social sentiment, prediction-market odds. It is a candidate signal being graded on the same after-fee bar as every other candidate. It never places orders and never overrides the hard rules. First live run, 2026-08-16: BTC-USD rated Underweight, target 55000, horizon 2-4 months (04-outputs/ta-reports/reports/BTC-USD_20260816_173215/).

D.2 Paper Trading

03-inputs/paper_trade.py simulates the book locally against live Binance public prices (no keys, no orders, not even testnet). State in 05-research/memory/paper-portfolio.json; every entry and exit journals to 05-research/memory/trade-log.md; 05-research/memory/portfolio.md is rewritten on every mark. The engine enforces the hard rules mechanically: max 3 new positions per rolling week, 5% notional cap (2% default), 10% cash floor, -7% hard stop from entry, 10% trailing stop from peak, never average down. Kelly inputs are journaled with each entry; with no signal past the after-fee bar, p is unvalidated, f* is at or below zero, and every paper entry is flagged as a rehearsal override rather than a bet Kelly endorses.

open -from-ratings opens only committee Buy or Overweight names; mark applies stops and rewrites the snapshot; a periodic mark loop turns the journal into a live paper track record.

D.3 Live Readiness

Live trading requires, in order: Binance API keys in the environment, `BINANCE_TESTNET=true` burn-in through `03-inputs/trade_binance.py`, and only then the deliberate `LIVE_TRADING=true` switch. The standing gate is unchanged: no signal, committee included, has yet beaten buy-and-hold and a coin flip on the out-of-sample after-fee scoreboard, so the honest book is small, defensive, and mostly cash until one does.